

Price Discovery in Bitcoin: The Role of Limit Orders

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Abstract

We analyse the price discovery process in bitcoin-dollar trading on Coinbase, the most established cryptocurrency spot exchange. Using a modified reinforcement learning framework, we find that market orders individually carry more information than limit orders, but due to their abundance, limit orders are the main driver of price discovery. Moreover, we observe a significant asymmetry in permanent price impact, with sell market orders carrying more information than buy market orders. We relate this asymmetry to unidirectional retail order flows. Owing to a negligible relative tick size, most of the price discovery contribution of limit orders originates from orders undercutting the best quotes. Consistent with theoretical predictions, the state of the order book has a significant impact on price discovery, with orders submitted along the order book imbalance showing an increased contribution and the importance of market orders being more pronounced in case of a binding bid-ask spread. Finally, the price discovery process on Coinbase is strongly influenced by information flows from other exchanges.

Keywords: Price Impact; Limit Order Book; Liquidity Provision; Price Discovery; Information Content; Reinforcement Learning.

JEL classification: C22, C5, E42, F31, G1, G2

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1 Introduction

Traditional microstructure models assume that informed agents incorporate their private information into prices by seeking immediacy and removing liquidity, whereas market makers are uninformed and their quotes reflect only public information (Kyle, 1985). While such a classification was reasonable in traditional markets where liquidity was provided through a designated market maker, it is overly simplistic nowadays. With most exchanges operating electronic limit order books where any participant is able to act as liquidity provider, the process of price discovery has fundamentally changed. Investors with an informational advantage have to choose between market and limit orders, where the decision might depend on the current state of the market, the expected behaviour of other market participants or even the time of day. In addition, the advent of algorithmic trading has tremendously increased both the number of messages sent to exchanges and the speed at which information is impounded into prices (O'Hara, 2015).

Despite their comparatively short history, cryptocurrencies have not been spared from this transformation in market microstructure. On the contrary, even the earliest crypto exchanges have been fully electronic, with trading occurring through a publicly visible limit order book. Modern cryptocurrency trading venues operate application programming interfaces (APIs) which allow users to connect their trading algorithms to the exchange without any intermediary and retrieve market data as well as manage their orders. Even less experienced retail traders can easily engage in algorithmic trading by using one of the countless available out-of-the-box trading bots.

In this study, we use an extensive tick-by-tick data set on transactions and order book messages on Coinbase, the most established and only listed cryptocurrency spot exchange, to analyse the price discovery process in bitcoin-dollar trading at its most granular level. In total, our data set covers the four-month period from February to May 2022 which, due to crypto exchanges operating 24/7, corresponds to 120 trading days. The data contain every message sent to the exchange, which allows us to identify each order submission and cancellation uniquely and eliminates the need to construct limit order flow variables as in Chaboud et al. (2021) for example. To be consistent with related studies such as Fleming et al. (2018), we first consider only market orders (trades) and examine how their permanent price impact is related to their size.¹ Then, to analyse the importance of limit order activity, we study all order flow components – market orders, limit order submissions, and limit order cancellations – jointly. In this context, we first quantify the price discovery contribution of limit order actions depending on their aggressiveness, before more closely examining the relationship between permanent price impact and order size. Several empirical studies have shown that in modern financial markets, the shape of the order book affects the behaviour of market participants and consequently, the price discovery process (Cao et al., 2009; Brogaard et al., 2014). Therefore, we then integrate the order book in our study and analyse how the price discovery contribution of the different order flow components changes depending on the prevailing order book imbalance and the quoted bid-ask spread. Finally, having closely analysed price discovery on Coinbase, we broaden our perspective and

¹We use the term market order for all marketable orders, i.e. orders that are immediately executed when submitted. Therefore, market orders also include buy (sell) limit orders where the limit price is greater (smaller) than the best ask (bid) quote.

take into account information flows from other exchanges. The crypto market is highly fragmented with hundreds of electronic exchanges, where the trading volume on some of them exceeds that on Coinbase by far. Therefore, it is likely that information from these exchanges influence the price discovery process on Coinbase. To analyse this, we study the permanent price impact of the different order flow components on Coinbase depending on the preceding return on other crypto derivatives and spot exchanges.

To analyse the permanent price impacts of different order flow components as well as their interaction, we rely on a modified reinforcement learning (RL) framework recently proposed by Philip (2020). Compared to the standard empirical tool for evaluating price discovery – the Vector Autoregression (VAR) model (Hasbrouck, 1991) – the RL framework offers two major advantages. First, unlike the VAR, the RL framework is able to capture nonlinear relationships. For example, many empirical studies report a concave relationship between price impact and trade size (Donier and Bonart, 2015). The traditional basic VAR model is not able to capture this relationship correctly and might lead to erroneous conclusions. In contrast, as shown by Philip (2020) and Kwan et al. (2021), the modified RL model is able to identify such nonlinear relationships. Second, the RL framework consistently outperforms the VAR in terms of computation time. By expanding the space of variables, the basic VAR model can be modified to capture nonlinearities. However, the estimation time increases exponentially in the number of coefficients. In contrast, the computational time required to estimate the modified RL model increases only linearly in the number of variables. Therefore, the RL framework is better suited to study permanent price impact, especially if multiple market variables are considered jointly.

Our analysis shows that market orders individually exhibit a much larger permanent price impact, and thus carry more information, than limit order actions. In particular, the individual information content of market orders is two to three times greater than that of undercutting limit order submissions. However, if the abundance of limit order actions is taken into account, limit order submissions are significantly more important than market orders and contribute about 50% to price discovery. Moreover, we find a significant asymmetry in the permanent price impact of market orders that persists throughout the entire sample period and has not been observed in traditional asset classes. Specifically, sell-initiated market orders contain 60% more information than orders originating from the buy side. We suppose that this asymmetry is caused by unidirectional retail order flows on Coinbase, where retail investors are mainly on the buy side of transactions. Since these investors are typically less informed than institutions, the large proportion of buy transactions initiated by retail traders reduces the information content, and consequently the permanent price impact, of buy-initiated market orders, causing an asymmetry.

For limit orders, the negligible relative tick size of less than one-third basis point makes submissions inside the spread highly attractive and thus, the bulk of the price discovery contribution of limit order submissions originates from undercutting orders. Cancellations of limit orders also contribute a significant amount to price discovery (34%) where, similar to submissions, most of this contribution comes from price-changing cancellations. However, for these cancellations, a large part of the permanent price impact is of mechanical nature. The extremely low tick size causes many empty price levels behind the best quotes, which in turn cause price-changing cancellations to trigger a rather large immediate

changes in the mid price.

Moreover, our results show that the state of the order book has a significant impact on the price discovery process. Trading actions whose direction coincides with the imbalance prevailing in the order book contribute considerably more to price discovery than actions opposite to the imbalance. Similarly, the quoted bid-ask spread affects the order choice of informed agents. If the spread is more than one tick, price discovery predominantly occurs through undercutting limit orders – their contribution is 57%. However, in case of a binding spread, market orders become the main driver of price discovery, contributing about 38%. Finally, we find that the price discovery process on Coinbase is heavily influenced by information flows from other exchanges. In particular, the price evolution of the three most actively traded bitcoin derivatives products and the largest bitcoin spot pair, within the second before a trading action on Coinbase determines the direction and magnitude of the permanent price impact of the trading action. In fact, the preceding price evolution on the three derivatives products and the largest spot pair is even more relevant for the price discovery process on Coinbase than the preceding price movement on Coinbase itself.

This study contributes to the existing literature in several ways. First, our results confirm the central assumption of modern microstructure models, such as [Rosu \(2020\)](#), that, in order to impound their information into prices, informed agents not only use market orders (informed liquidity taking) but also rely heavily on limit orders (informed liquidity making). Similarly, the results also support recent models of limit order book markets such as [Bhattacharya and Saar \(2020\)](#) and [Riccó et al. \(2022\)](#) in their empirical predictions that order book variables contain significant amounts of information and influence the price discovery process. Second, our study contributes to the empirical literature on price formation in modern financial markets where trading occurs through central limit order books. While the theoretical market microstructure literature has recognized the fundamental shift from informed liquidity taking to informed liquidity provision and has started to incorporate it into models, the number of detailed empirical studies is still limited, mainly due to the lack of available data. Existing works focus on US Treasury securities ([Fleming et al., 2018](#)), Canadian equities ([Brogaard et al., 2019](#)) and FX markets ([Chaboud et al., 2021](#)). We expand this spectrum to include the relatively new asset class of cryptocurrencies. Even though the crypto market has experienced an increasing institutional adoption and has recently moved once more into the focus of regulators, its microstructure is still little researched and not yet completely understood.² While there are several works such as [Alexander and Heck \(2020\)](#) that quantify inter-exchange price discovery, mostly relying exclusively on transaction data, we are the first to provide insights into the most fundamental aspects of the intra-exchange price discovery process, by considering not only transaction data, but all messages sent to the exchange. In particular, we show that even within the still young and comparatively small cryptocurrency market, liquidity providers are well informed and their actions are highly relevant in terms of price discovery.

The remainder of this paper is organized as follows: Section 2 explains the methodology of the modified reinforcement learning framework used to measure permanent price impact. Section 3 describes the cryptocurrency trading landscape as well as the data set used in this study. Section 4 presents our results on permanent price impacts and price discovery contribution of market orders, limit order

²See [Financial Times](#) on 24 June 2022.

submissions and cancellations. Sections 5 and 6 explore how the price discovery process on Coinbase is influenced by the state of the order book and the price evolution on other crypto trading venues, respectively. Section 7 summarises and concludes.

2 Methodology

In this section, we introduce the main idea and the most important features of the modified reinforcement learning (RL) framework used to quantify permanent price impact. For more details on RL itself and the augmentation of the original RL setting as well as for very illustrative examples, we refer to Philip (2020). In terms of assumptions, the RL approach is minimal and only assumes that the data are stationary and Markov. Compared to the traditional VAR model, which assumes linearity, these conditions are quite unrestrictive. Within the bitcoin market, Ghysels and Nguyen (2019) find no evidence for non-Markovian learning, suggesting that the Markovian learning assumption typically adopted in limit order book models is reasonable.

In the RL framework proposed by Philip (2020), traders interact with the market in discrete time steps. Formally, at each time step, the market can be in a certain state $s_i \in S$ where S is called the state space. This space is very general and can be defined depending on the research question. In the most simple case, there is only one state (i.e. open), while more sophisticated states could for example capture the current order book imbalance or the available liquidity. At each time step, market participants are allowed to perform certain actions $a_j \in A$ with A denoting the action space. As for the market states, this set of available actions can be modified depending on the object of the analysis. Practical examples are the submissions of market orders or the cancellations of limit orders. In the following, we denote the state of the market at time t by $s(t)$ and the action that is performed at time t by $a(t)$. The model can operate either in event time or in calendar time. We choose to use the former since this reduces the bias arising from irregularly spaced observations (O'Hara, 2015).

The actions represent interactions of agents with the market that might change its state and trigger reactive actions by other agents in the next time step. Specifically, if at time t , an agent performs action $a(t)$, the market might transit from its current state $s(t)$ to some state $s(t+1)$ at time $t+1$. At this next time step, where the market is in the new state $s(t+1)$, other agents – or even the same agent that initiated action $a(t)$ previously – might then take some action $a(t+1)$. For example, if the bid-ask spread is at its minimum value, the submission of a very large buy market order could absorb all liquidity available at the first few ask levels and consequently widen the spread. Due to this widening, market makers might then submit sell limit orders to restore the removed liquidity.

Due to this interaction between agents and the market, each state-action pair $(s_i, a_j) \in S \times A$ can be assigned a probability to trigger a transition to the state-action pair (s_k, a_l) . Formally, this is the probability that if the market is in state s_i at time t (i.e. $s(t) = s_i$) and an agent performs action a_j (i.e. $a(t) = a_j$), the market will be in state s_k at the next time step $t+1$ (i.e. $s(t+1) = s_k$) where then action a_l is performed (i.e. $a(t+1) = a_l$).³ Following Philip (2020), these transition probabilities

³The state-action pair (s_i, a_j) can also trigger a "transition" to itself. In fact, financial markets typically exhibit a diagonal effect, where the probability to observe a given action is increased after this action has just occurred (Biais et al., 1995).

can be empirically estimated by maximum likelihood estimation, as

$$\begin{aligned} P_{(i,j),(k,l)} &:= P(s(t+1) = s_k, a(t+1) = a_l \mid s(t) = s_i, a(t) = a_j) \\ &= \frac{N(s_k, a_l \mid s_i, a_j)}{\sum_{s_k \in S} \sum_{a_l \in A} N(s_k, a_l \mid s_i, a_j)} \end{aligned} \quad (1)$$

where $N(s_k, a_l \mid s_i, a_j)$ denotes the number of times the state-action pair (s_i, a_j) is followed by state-action pair (s_k, a_l) . Intuitively, the probability in (1) is simply the number of times (s_i, a_j) is followed by (s_k, a_l) , relative to the number of times (s_i, a_j) occurs in the data set.

Besides the ability to change the market state, actions might also cause an immediate change in the mid price where the magnitude of this change might further depend on the current market state. For instance, a very large buy market order leads to a larger immediate change in the mid price if the depth at the first ask levels is low. Therefore, each state-action pair (s_i, a_j) exhibits an immediate impact on the mid price, denoted by $R(s_i, a_j)$. Note that this immediate impact is of purely mechanical nature and for some actions, such as limit order submissions behind the best quotes, it will always be zero. Once more following Philip (2020), we estimate $R(s_i, a_j)$ as the average demeaned log change in the mid price observed for the state-action pair (s_i, a_j) .

The permanent price impact of (s_i, a_j) is the long-run change in the mid price caused by the state-action pair, i.e. by an agent taking action a_j when the market is in state s_i . As explained above, there are two consequences to the occurrence of a state-action pair. It causes an immediate change in the mid price and triggers a transition to some other (or possibly the same) state-action pair in the next time step. Therefore, both of these consequences have to be taken into account when calculating the permanent price impact. Philip (2020) does this by measuring the permanent price impact $Q(s_i, a_j)$ of (s_i, a_j) recursively as its immediate impact $R(s_i, a_j)$ plus the discounted expected permanent price impacts of all subsequent state-action pairs (s_k, a_l) triggered by the initial state-action pair (s_i, a_j) . Put in mathematical terms, this yields the value iteration rule

$$Q(s_i, a_j) = R(s_i, a_j) + \gamma \sum_{k=1}^{n(S)} \sum_{l=1}^{n(A)} P_{(i,j),(k,l)} Q(s_k, a_l) \quad (2)$$

where $n(S)$ and $n(A)$ denote the cardinality of S and A respectively, and γ represents a discount factor that in a high-frequency analysis is typically set very close to 1. Such a value iteration update is commonly known as Q -learning in the RL literature. Watkins and Dayan (1992) show that the iteration of Equation (2) will converge to the optimum with probability one so long as all actions are repeatedly sampled in all states and the action-values are represented discretely. Given that these requirements are met and both the immediate impacts and the transition probabilities are calculated as explained above, we can estimate the permanent price impact $Q(s_i, a_j)$ of (s_i, a_j) by initializing it as some constant (usually zero) and then iterating Equation (2) until convergence.

The transition probabilities and the immediate impacts are empirically estimated input parameters and thus, they exhibit a certain finite-sample error, which can cause estimation errors in the permanent price impact estimates. Therefore, to quantify these estimation errors, we follow the block bootstrap-

ping procedure in Philip (2020). Specifically, a single bootstrap iteration involves the following steps. First, we decompose the original data sample of size n into blocks of l consecutive observations. Then, we sample n/l blocks with replacement from this set of blocks and concatenate the sampled blocks, so that we again obtain a sample of the same size as the original data. For this bootstrapped sample, we can then compute the permanent price impacts. We repeat this procedure 100 times and calculate confidence intervals over the 100 bootstrapped estimates of permanent price impact. Following White and White (2010), the block length l is chosen as $n^{1/3}$.

Finally, in a similar spirit to Hasbrouck (1991), Philip (2020) also introduces a simple metric which assigns to each state-action pair its relative contribution to the total price variation in the model. The metric is calculated as

$$RC(s_i, a_j) = \frac{|Q(s_i, a_j)| \times N(s_i, a_j)}{\sum_{k=1}^{n(S)} \sum_{l=1}^{n(A)} |Q(s_k, a_l)| \times N(s_k, a_l)} \quad (3)$$

where $N(s_i, a_j)$ denotes the number of times the state-action pair (s_i, a_j) occurs in the data set. This metric is highly flexible and can easily be modified to obtain the contribution of an action to the total amount of price variation within a certain market state or even the contribution of a state to the total price variation.

3 Data

Our analysis is based on an extensive tick-by-tick data set obtained from [Tardis.dev](https://tardis.dev). In particular, we obtain data on all transactions and order book messages on Coinbase during the period from 1 February to 31 May 2022. Since crypto markets operate 24/7, this sample period corresponds to 120 trading days. The transaction records specify the transaction price, the amount that was traded, as well as the order sign, i.e. a flag indicating the taker side of the transaction. If a marketable order executes against multiple resting limit orders, the transaction records contain a separate entry for each of the limit orders. However, according to Upson et al. (2018), interpreting these entries as separate transactions introduces significant bias into empirical studies. Therefore, we combine consecutive trade records with the same order sign and the exact same timestamp into a single transaction.

The order book data set is split so that we obtain a separate file for each trading day in our sample. Each of these 120 files starts with an initial snapshot of the entire order book, containing all active price levels and the respective available volumes. After this snapshot, all updates to any of the price levels are recorded.⁴ Specifically, for each change in the order book, we obtain a record specifying the price level that was changed, the order book side of the change, whether the volume at this level was increased or reduced, as well as the volume available at this price level after the change.

Both types of data are double timestamped with microsecond precision, once by the exchange and a second time when the message arrives at the Tardis servers. Importantly, for the transaction records,

⁴Due to latency issues, the initial snapshot is usually not taken exactly at midnight but a few milliseconds after. Therefore, it happens that on some days, the file contains a few order book update messages before the initial snapshot. Since we cannot accurately determine the top-of-book quotes for these updates without going back to data from the previous day, we ignore these few messages.

Table 1 Summary Statistics: Market and Limit Order Activity

The table reports averages on market and limit order activity in BTCUSD on Coinbase during the period from 1 February to 31 May 2022. The ratio of average to standard deviation is shown in parentheses. *Count* represents the daily number of observations for each action. *Size* denotes the order size, either in USD or in BTC. *Duration* denotes the average time between two actions of the same type (in milliseconds).

		Daily Count	Size (in BTC)	Size (in USD)	Duration
Trades	Buy	202,862 (3.24)	0.0448 (0.16)	1,680.07 (0.16)	425.91 (0.70)
	Sell	117,386 (2.46)	0.0725 (0.17)	2,728.05 (0.17)	736.03 (0.57)
Limit Order Submissions	Buy	6,312,348 (3.72)	0.2474 (0.00)	7,865.76 (0.27)	13.69 (0.04)
	Sell	6,023,769 (3.58)	0.2144 (0.21)	8,702.35 (0.01)	14.34 (0.04)
Limit Order Cancellations	Buy	6,140,126 (3.81)	0.2489 (0.00)	8,034.19 (0.27)	14.07 (0.04)
	Sell	5,845,214 (3.67)	0.2194 (0.22)	8,913.68 (0.01)	14.78 (0.04)

the exchange-provided timestamp indicates the exact time at which the market order causing the trade hit the order book. This eliminates time delay issues as in [Hautsch and Huang \(2012\)](#) and allows us to match transactions and order book updates on this exchange-provided timestamp so that we can uniquely identify the volume reductions in the order book that are caused by market orders, rather than cancellations of limit orders.

Trading on major cryptocurrency exchanges is very active and fast-paced. Over our 120-day sample period, there were more than 2.9 billion order book changes and almost 39 million transactions. For comparison, [Fleming et al. \(2018\)](#) report 2.4 billion order book messages for US Treasury securities over 500 trading days in 2010 and 2011, where – similar to crypto exchanges – trading takes place almost 24 hours per day. Table 1 reports summary statistics on the market and limit order activity over our sample period from 1 February to 31 May 2022. We see that limit order actions occur much more frequently than market orders. With a daily average of about 24 million limit order messages and only 320,000 transactions, limit order activity is responsible for almost 99% of all messages sent to the exchange. Highlighting the fast paced nature of cryptocurrency exchanges, submissions and cancellations of limit orders occur on average every 14 milliseconds. The number of submissions and cancellations is almost equal, with about 97% of all limit orders being cancelled. Such high cancellation rates are not unique to cryptocurrencies but they are a common characteristic of modern financial markets ([O’Hara, 2015](#)). Moreover, transactions exhibit two statistically significant asymmetries. While trades are more often initiated from the bid side than from the ask side, the average size of sell transactions is about 60% greater than that of buy-initiated trades. Finally, we note that the size of limit order submissions and cancellations is about four times that of transactions.

Table 2 provides summary statistics on the quoted bid-ask spread and the absolute price distances among the top five levels in the order book, based on one-second snapshots. We see that the bid-ask spread is binding for about 37% of the time. However, with a standard deviation of more than two dollars, it is highly volatile and sometimes reaches even values above 150 USD, which highlights the high volatility of bitcoin ([Scaillet et al., 2020](#)). Despite these extreme values, the bid-ask spread is in relative terms still very small, with its average value of 1.73 USD consistently corresponding to less than a basis point of the BTCUSD price during our sample period. The statistics on the distances between adjacent price levels behind the best quotes indicate a large number of empty price levels

Table 2 Summary Statistics: Spread and Price Distance

The table reports summary statistics on the bid-ask spread and the absolute price distances among the top five levels in the order book. *Binding* represents the proportion of observations (in %) where the variables are at their minimum value of 0.01 USD. All statistics are calculated based on one-second order book snapshots during the period from 1 February 31 May 2022. Except *Binding*, all quantities are in USD.

	Spread	Bid 1-2	Bid 2-3	Bid 3-4	Bid 4-5	Ask 1-2	Ask 2-3	Ask 3-4	Ask 4-5
Average	1.73	0.86	0.71	0.62	0.59	0.80	0.66	0.59	0.55
St. Dev.	2.13	1.38	1.12	0.97	0.88	1.28	1.03	0.91	0.83
Min.	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Median	1.04	0.15	0.19	0.20	0.22	0.18	0.21	0.21	0.22
Max.	196.02	186.95	95.53	107.28	107.28	126.24	91.34	126.17	77.25
Binding	37.39	36.45	29.68	27.09	23.53	34.17	27.61	25.93	23.15

in the order book. For instance, the absolute distance between the best and second best quote is on average 0.80 USD on the ask side and 0.86 USD on the bid side. These empty price levels are caused by the tick size of 0.01 USD, which in relative terms, corresponds to only about 0.0025 bps of the bitcoin price. This negligible relative tick size makes bitcoin similar to (very) small-tick stocks for which the limit order book is usually sparse and exhibits many empty price levels.

4 Permanent Price Impact

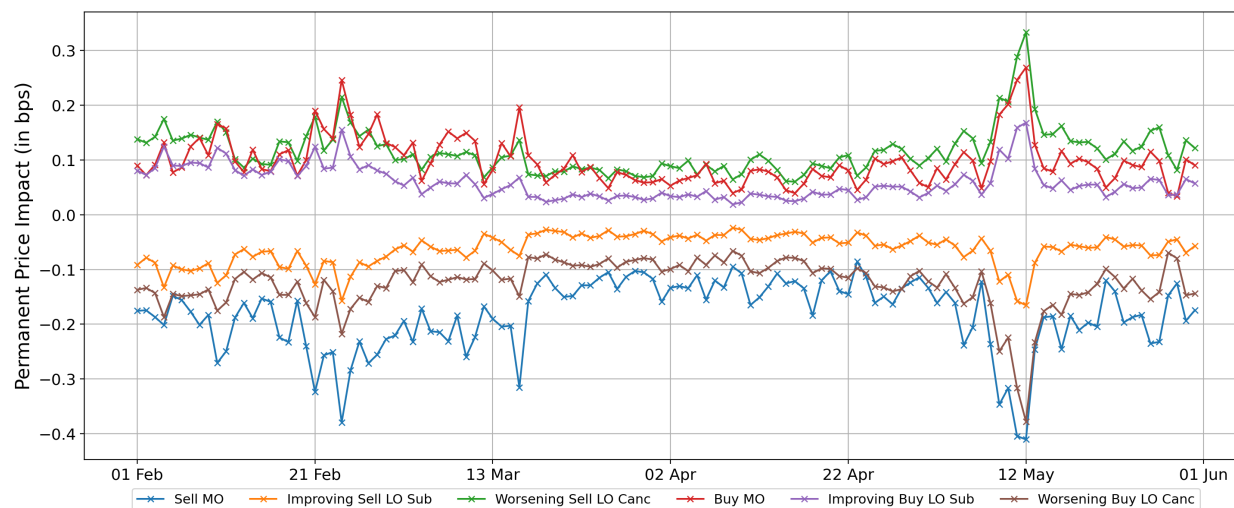
In this section, we study in detail permanent price impacts and price discovery contributions of all order flow components – market orders, limit order submissions and limit order cancellations. However, for now, we ignore the state of the order book that prevails when the respective action occurs. That is, we do not take into account, for example, whether the order book is imbalanced or whether the bid-ask spread is narrow. Therefore, throughout this section, we assume in the RL model that there is only one possible market state (i.e. open). This way, we obtain price impact estimates that are averaged over all possible states of the order book. In Section 5, we will then take into account different market states and analyse their influence on price impact.

Due to the sheer size of our order book data, it is computationally not feasible for us to train the RL model and compute block-bootstrapped confidence intervals based on the entire sample period from 1 February to 31 May 2022. Therefore, we randomly choose the week from 18 to 24 April as a representative interval in our sample period for which results are presented in this section. Limiting the timeframe to this seven-day period allows us to train the model in a reasonable amount of time, while still leading to very tight confidence intervals and reducing the impact of possible day-of-the-week effects. To justify our procedure, Figure 1 depicts the evolution of permanent price impact over our sample period, that is, we train the RL model and estimate price impact separately for each of the 120 days in our sample period. We see that the price impact of all trading actions is significantly elevated during the high-volatility regimes from mid-February to mid-March and the beginning of May. Overall however, the permanent price impact estimates are quite consistent and the ranking of individual actions by the magnitude of their impacts varies very little, substantiating our procedure of focussing on a representative week.

Figure 2 depicts the evolution of the BTCUSD mid price on Coinbase between 18 and 24 April

Figure 1 Permanent Price Impact Over Time

The figure displays the permanent price impacts (in basis points), estimated separately for each day in our sample period from 1 February to 31 May 2022. To increase visibility, we only show the impacts of market orders (MO), price-improving limit order submissions (LO Sub), and price-worsening limit order cancellations (LO Canc).



2022. After a temporary initial drop in the morning of 18 April from almost 40,000 USD to below 39,000 USD, the price steadily increased and peaked at slightly below 43,000 USD in the afternoon of 21 April. After that, it dropped again to about 39,500 USD during the afternoon of 22 April and remained at this level until the end of the week.

The earliest microstructure models such as [Kyle \(1985\)](#) suggest that only trades carry information. Therefore, we start our analysis by quantifying the permanent price impact of market orders in Section 4.1. Initially, we consider only the order sign of trades in our model. In a second analysis, we then differentiate market orders based on their size to examine the relationship between permanent price impact and trade size. In Section 4.2, we then incorporate limit order activity and train the model on all order flow components – market orders, limit order submissions, and limit order cancellations – jointly. Since limit orders are more nuanced than market orders, we first train a model including order sign and aggressiveness. Then, we additionally differentiate all orders based on their size. Finally, in Section 4.3, we analyse the intraday pattern of permanent price impact and price discovery on Coinbase.⁵

4.1 Market Orders

We start by analysing the price impact of market orders (trades). To this end, we train the RL model with the most simple action space – submitting a buy market order and submitting a sell market order. In contrast to the traditional basic VAR used for example in [Chaboud et al. \(2021\)](#), the RL approach is able to distinguish between buy and sell market orders and therefore, it yields different estimates for

⁵Due to the very large sample size, the confidence intervals of all estimates are very tight and therefore, to increase visibility, we do not show them in the figures of this paper.

Figure 2 BTCUSD Mid Price

The figure displays the second-by-second mid price of BTCUSD on Coinbase from 18 to 24 April 2022.



the two types of orders.⁶ Table 3 reports the permanent price impact estimates for buy and sell market orders (in basis points) together with block-bootstrapped confidence intervals and standard errors, as well as the number of observations for each action. We observe a statistically significant asymmetry, with sell market orders having a higher price impact (in absolute terms), and consequently carrying more information, than buy market orders. While sell-initiated orders reduce the mid price by about 0.14 basis points, orders originating from the buy side lead to an increase of only 0.09 bps in the mid price. Figure 1 shows that this asymmetry is not unique to the representative week from 18 to 24 April, but it persists throughout the entire sample period.

Such an asymmetry is not reported in other price discovery studies. In fact, [Fleming et al. \(2018\)](#) even show that for US Treasury securities, permanent price impact is not significantly different for buy and sell trades. We suppose that the asymmetry in Table 3 is related to the investor structure in the cryptocurrency space. In most asset classes, retail investors play a minor role. For example, in US equity markets, retail trading is responsible for less than 20% of the total trading volume.⁷ In crypto markets however, retail investors take on a significant role, with their trading accounting for one third of the total trading volume on Coinbase.⁸ Moreover, in contrast to traditional asset classes, smaller retail investors have repeatedly triggered strong bubble-like price increases in the cryptocurrency space.⁹ Therefore, we suppose that the observed asymmetry in price impact is caused by highly unidirectional retail order flow on Coinbase. Retail investors have a rather positive view on the future of bitcoin – or are possibly even driven by the fear of missing out on large profits – and thus, they prefer to buy, rather than sell, bitcoin. Since this type of investors is usually less informed than institutions,

⁶The VAR model can be expanded to distinguish between buy and sell market orders as in ([Fleming et al., 2018](#)) but this increases the number of coefficients, and thus the estimation time, exponentially ([Philip, 2020](#)).

⁷See [Bloomberg](#) on 17 November 2021.

⁸See [Financial Times](#) on 13 May 2022. Since Coinbase is the most established cryptocurrency exchange and thus quite attractive for institutions, the retail trading volume is likely to be higher on other unregulated exchanges.

⁹See [Financial Times](#) on 28 June 2022.

Table 3 Permanent Price Impact: Market Orders – Trade Sign

The table reports the permanent price impact estimates (in basis points) for market orders, together with bootstrapped confidence intervals and standard errors, as well as the number of observations for each action. The confidence intervals and standard errors are computed from 100 simulations following the block bootstrapping procedure described in Philip (2020).

Action	Count	Price Impact	Lower 2.5%	Upper 2.5%	St. Err.
Sell	792,735	-0.1419	-0.1434	-0.1407	0.0007
Buy	1,295,979	0.0868	0.0859	0.0879	0.0005

the large proportion of buy transactions initiated by retail traders reduces the information content, and consequently the permanent price impact, of buy-initiated market orders, causing the asymmetry documented in Table 3. Such a hypothesis is very difficult to confirm empirically since only proprietary exchange data would allow us to conduct appropriate tests. However, the summary statistics on trade sizes in Table 1 lend some credence to our hypothesis. While there are significantly more market orders originating from the bid side than from the ask side, the average size of sell-initiated market orders is more than 60% larger than that of buy-initiated orders.¹⁰ Since small trades are usually considered a proxy for retail trading (Han and Kumar, 2013), this significant difference in trade size supports our hypothesis that the asymmetry in permanent price impact of buy and sell market orders is caused by retail traders being mainly on the buy side of the transaction.

Compared to Brogaard et al. (2019) and Philip (2020) who study price impact on Canadian and Australian equity markets, respectively, and document impacts between -3.06 bps and 3.25 bps, the order of magnitude of the permanent price impacts in Table 3 is very small. The reasons for this difference are trading frequency and tick size. As documented by Hautsch and Huang (2012), the strength of market responses is negatively correlated to trading frequency and positively influenced by relative tick size. Comparing the trading frequency of Australian and Canadian equities to that of BTCUSD on Coinbase, we find a massive difference. The most traded stock in Philip (2020) exhibits about 280,000 trades during a two-month period, while the average daily number of transactions for Canadian stocks included in Brogaard et al. (2019) is less than 5,000. In contrast, our dataset on BTCUSD includes more than two million transactions within the representative week from 18 to 24 April. Moreover, Coinbase has a tick size of 0.01 USD which in relative terms corresponds to only about 0.002 to 0.003 bps of the BTCUSD price. In contrast, Australian and Canadian equities included in Brogaard et al. (2019) and Philip (2020) typically exhibit a relative tick size of more than one basis point. Following Hautsch and Huang (2012), both of these features – the high number of transactions and the negligible relative tick size – contribute to the small magnitude of permanent price impacts in BTCUSD. In fact, the price impact estimates in Table 3 are of a similar magnitude as those in Chaboud et al. (2021) who document an average impact of about 0.1 bps for market orders in euro-dollar trading on Electronic Broking Services (EBS) during the period from 2008 to 2017.

It is well documented that the price impact of a trade is positively correlated with its size, with many studies reporting a concave functional relationship between trade size and price impact (Cont et al., 2014; Philip, 2020). However, within the crypto market, little to no research has been conducted to

¹⁰Note that this size difference between buy and sell transactions does not explain the asymmetry in Table 3. The next analysis shows that a sell trade has a significantly larger impact than a buy trade of the same size.

investigate this issue. The only notable study dealing with price impact within crypto markets is [Donier and Bonart \(2015\)](#) who analyse the price impact of meta orders on the now-defunct cryptocurrency exchange Mt. Gox, confirming the well-documented square-root relationship between price impact and trade size for meta orders. It is worth mentioning however that (i) this study focusses on the impact of meta orders and (ii) at the time of the study, the crypto market was still in its infancy and has transformed significantly since then. Therefore, we examine the relationship between permanent price impact and trade size within today's crypto market more closely.

To incorporate trade sizes, we expand the action space of the previous RL model by quantizing the size of market orders. Specifically, we divide orders into six quantile groups depending on their size and define submissions of orders in each group as a separate action. This additional differentiation expands the action space of the RL model from two to twelve actions (six quantile groups for buy orders and six quantile groups for sell orders).¹¹ Table 4 reports the resulting price impact estimates, confidence intervals, standard errors as well as the number of observations for each action. We see that permanent price impact is generally increasing in signed trade size. For example, a very small buy trade (Q1) exhibits a price impact of 0.05 bps, whereas a very large buy transaction (Q6) moves the mid price by 0.30 bps. Similarly, the smallest sell trades (Q1) lead to a decrease of 0.09 bps in the mid price, while large sell-initiated trades (Q6) exhibit an impact of 0.32 bps. Moreover, we see that the asymmetry in permanent price impact between buy and sell transactions that we already documented in Table 3 persists across all order size groups. That is, sell-initiated market orders have a higher absolute price impact, and thus contain more information, than buy market orders of the same size. As explained above, we suppose that this asymmetry is related to retail investors being mainly on the buy side of transactions. Further supporting this hypothesis, Table 4 shows that in the first five order size groups, there are significantly more buy-initiated transactions, while in the top quantile group, sell trades predominate. Finally, it is worth mentioning that permanent price impact is not monotonically increasing in signed order size. Rather small market orders (Q2) exhibit a lower absolute impact than very small trades in the first quantile size group. This artefact seems however to be related to the interaction between trades and limit orders, since it disappears (or is at least greatly reduced) once we incorporate limit order activity in the RL model in Section 4.2.

Figure 3 displays the permanent price impact of buy and sell trades as a function of their signed size, i.e. sell-initiated market orders are depicted by negative trade sizes. It is clearly visible that the relationship between permanent price impact and trade size is not linear, but rather concave. That is, price impact is overall increasing with trade size, but the rate of increase reduces as trade size grows, which implies that small trades exhibit in relative terms a lower impact than large transactions. This finding is consistent with [Philip \(2020\)](#) and confirms their conclusion that the basic linear VAR with trade size as measure for market order flow might lead to erroneous conclusions about permanent price impact.

¹¹For consistency and comparability, the quantile groups are computed across all trades, instead of separately for buy and sell transactions.

Table 4 Permanent Price Impact: Market Orders – Trade Size

The table reports the permanent price impact estimates (in basis points) for market orders, depending on their size. Orders are divided into quantile groups (Q1 to Q6) depending on their size. The confidence intervals and standard errors are computed from 100 simulations following the block bootstrapping procedure described in Philip (2020).

Action	Size	Count	Price Impact	Lower 2.5%	Upper 2.5%	St. Err.
Sell	Q1	115,554	-0.0921	-0.0937	-0.0902	0.0009
	Q2	103,422	-0.0664	-0.0678	-0.0649	0.0008
	Q3	135,293	-0.1249	-0.1271	-0.1227	0.0011
	Q4	111,772	-0.1005	-0.1023	-0.0989	0.0009
	Q5	148,347	-0.1469	-0.1490	-0.1449	0.0011
	Q6	178,347	-0.3169	-0.3212	-0.3127	0.0023
Buy	Q1	232,565	0.0469	0.0460	0.0481	0.0005
	Q2	244,697	0.0423	0.0413	0.0435	0.0006
	Q3	212,826	0.0710	0.0695	0.0725	0.0008
	Q4	236,347	0.0590	0.0580	0.0601	0.0006
	Q5	199,772	0.1138	0.1122	0.1152	0.0008
	Q6	169,772	0.2997	0.2953	0.3042	0.0024

4.2 Limit Order Activity

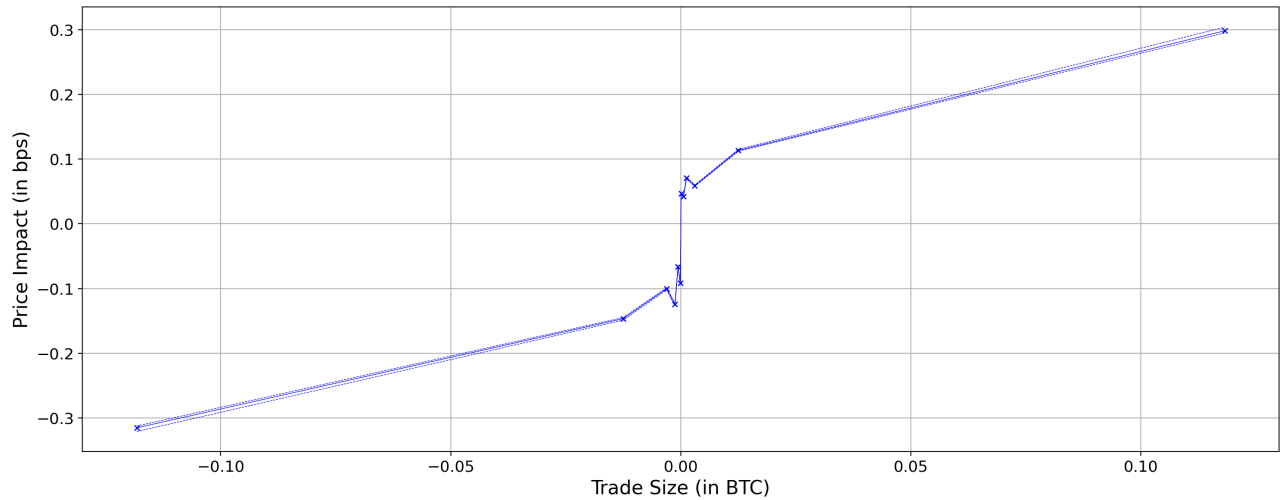
The earliest microstructure models such as Kyle (1985) assume that only trades contain and transmit information. However, since most modern markets operate a limit order book where all market participants can provide liquidity, this assumption is not reasonable anymore. Therefore, more recent models such as Rosu (2020) also allow limit orders to carry information. Recent studies empirically support this new approach to information transmission within financial markets (Brogaard et al., 2019; Chaboud et al., 2021). However, nothing is yet known about the role of limit orders within the crypto market. We fill this gap by providing detailed insights on the importance of limit order activity in terms of information transmission.

In contrast to the previous section, we now include all order flow components (market orders, limit order submissions, limit order cancellations) jointly. However, compared to market orders, limit order activity is more nuanced. Due the additional dimension of the limit price, a classification of limit orders solely by their sign would be far too general to assess their information content accurately. Therefore, we additionally differentiate both submissions and cancellations based on their aggressiveness, measured as the signed distance to the best quote on the same side of the order book. More specifically, we apply the following classification to limit order activity. Submissions are divided into ten predefined bins depending on their distance to the best same-side quote.¹² The exact definitions of the bins can be seen in Table 5. Here, the first bin captures submissions with a negative distance to the best quote, which corresponds to price-improving submissions. Limit order cancellations are divided into the same predefined bins, but since they can by definition not occur at a price better than the best quote, the first distance-to-best bin is redundant and can be removed, reducing the number of bins to nine. Nevertheless, a cancellation can still lead to an immediate change in the mid price, given

¹²Previously, we divided trade sizes into quantile groups. While this classification is still possible for the distance to the best same-side quote, we prefer to divide limit order submissions in predefined bins rather than quantile groups, since the vast majority of limit order activity takes place around the best quotes and therefore the distribution of the distance to the best same-side quote is heavily right-skewed.

Figure 3 Permanent Price Impact of Market Orders

The figure displays the permanent price impact estimates (in basis points) for market orders as a function of signed order size. Note that sell market orders are reported with negative trade sizes.



that it occurs at the best quote and there is no other limit order resting at this price. According to [Brogaard et al. \(2019\)](#), such price-worsening cancellations exhibit significantly different price impacts. Therefore, to capture this difference, we split cancellations occurring at the best quotes further, depending on whether they are price-worsening (i.e. their immediate impact is not zero) or immediately leave the best quote unchanged (i.e. zero immediate impact).¹³ As in the previous section, market orders are classified solely based on their sign.¹⁴ Table 5 illustrates the 42 actions resulting from this differentiation of market and limit order activity.

Having defined the action space properly, we can now train the RL model. The resulting estimates of permanent price impact are reported in Table 6. In addition, it displays the relative contribution of each action to the total amount of price variation, as defined in Equation (3). We see that market orders individually carry much more information than limit order submissions, even in case the latter improve on the best quotes. Specifically, buy (sell) market orders move the mid price by 0.06 bps (−0.11 bps) and thus carry two to three times as much information as submissions of limit orders inside the spread. Compared to limit order cancellations however, the dominance of market orders is not that pronounced. The impact of a price-worsening buy limit order cancellation is about −0.1 bps and therefore, more than 90% of the impact of a sell-initiated market order.¹⁵ On the ask side, the

¹³Note that for submissions, such a differentiation by immediate impact is already contained in the distance-to-best classification since submissions with a negative distance to the best same-side quote are price-improving and will by definition immediately impact the mid price. Also, limit order actions behind the best quotes will not immediately change the best quote and thus, they always exhibit a zero immediate impact.

¹⁴As in [Brogaard et al. \(2019\)](#), it would also be possible to differentiate market orders further based on whether they consume all liquidity available on the best quote. However, in order to stay consistent with [Chaboud et al. \(2021\)](#) and to not distort results too much by liquidity effects, we decided to include only the order sign as differentiating factor. As robustness check, we also trained the RL model where market orders are differentiated based on whether they consume all available liquidity, but the results – especially the price discovery contributions – are qualitatively very similar to those presented here.

¹⁵We compare buy (sell) limit order cancellations to sell-initiated (buy-initiated) market orders since their trading

Table 5 RL Model – Action Space

The table illustrates the classification of market and limit order activity. Market orders (MO) are differentiated based on order sign, limit order submissions (LO Sub) and limit order cancellations (LO Canc) are classified by order sign and signed distance to the best same-side quote (DTB; in USD). In addition, cancellations at the best quotes are differentiated based on whether they lead to an immediate change in the mid price (Worsening) or leave it unchanged (Matching). Note that the submission of a limit order with negative distance to the best quote improves the mid price by definition.

Criterion	Possible Values		MO	LO Sub	LO Canc
Order Sign	Buy		✓	✓	✓
	Sell		✓	✓	✓
DTB	< 0	Improving		✓	
	0	Worsening			✓
	0	Matching		✓	✓
	0.01			✓	✓
	0.02–0.05			✓	✓
	0.05–0.10			✓	✓
	0.10–0.25			✓	✓
	0.25–0.50			✓	✓
	0.50–1.00			✓	✓
	1.00–2.00			✓	✓
	2.00–3.00			✓	✓
	≥ 2			✓	✓

price impact of 0.09 bps is even larger than that of buy market orders. However, this large impact of cancellations seems to be of rather mechanical nature. The extremely small relative tick size of less than 0.003 bps on Coinbase leads to a very high granularity of the order book, which in turn leads to many empty price levels and thus, to often quite large price gaps behind the best quotes. According to Table 2, there are on average 80 empty price levels between the best and second best quote. These gaps then cause price-worsening cancellations at the best quotes to have a comparatively large immediate price impact. Specifically, during the representative week from 18 to 24 April, price-worsening cancellations exhibit an immediate impact of 0.086 bps for sell orders and -0.095 bps for buy orders, which is more than 96% of the the permanent price impact reported in Table 6. Therefore, a large portion of the high permanent price impact of cancellations can be attributed to a high immediate impact caused by the extremely small relative tick size.¹⁶

The most important takeaway from Table 6 is however that since limit order actions are far more numerous, the contribution of market orders to the total price variation is only about 16% and therefore much lower than that of limit order submissions (50%) and cancellations (34%). In total, limit order actions changing the best quotes (price-increasing submissions and price-worsening cancellations) contribute more than 71% to the total price variation and therefore, they are clearly the most important order flow instrument with regards to price discovery. More specifically, of that 50% contributed by limit order submissions, 87% comes from orders undercutting the best quotes, while price-worsening cancellations are responsible for 82% of the contribution caused by limit order cancellations (34%). In

directions coincide.

¹⁶Since we do not differentiate market orders based on whether they consume all liquidity available at the opposite-side best quote, the influence of this mechanical feature is reduced. For limit order submissions on the other hand, price gaps behind the best quotes are not relevant and the majority of undercutting limit orders are placed just one tick above the best bid or below the best ask, leading to an immediate impact of 0.01 USD.

general, this finding is in line with recent empirical studies including [Collin-Dufresne and Fos \(2015\)](#), [Fleming et al. \(2018\)](#), [Brogaard et al. \(2019\)](#), and [Chaboud et al. \(2021\)](#).

There are two cost-related reasons for the pronounced importance of (price-improving) limit order submissions with regards to information transmission. The first one is trading costs. Fees on Coinbase follow a tiered maker-taker model. Depending on the market participant's 30-day trading volume, liquidity-taking orders face fees ranging from 60 bps to 5 bps, while orders that add liquidity to the book are charged at most only 40 bps and for highest-volume traders, they are even free of charge.¹⁷ This fee structure naturally makes submitting limit orders more attractive compared to market orders: in order to overcome transaction costs, the trader's informational advantage needs to be significantly greater for market orders than for limit orders. The second reason for informed investors to use price-improving limit orders is the tick size. As already mentioned above, Coinbase has a tick size of 0.01 USD which in relative terms corresponds to only about 0.0025 bps, leading to an extreme granularity of the order book. This granularity allows market participants to undercut limit orders of other traders, and consequently gain price priority, at very low costs, which reduces the importance of submissions at the best quotes and makes price-improving limit order submissions highly attractive ([Dyhrberg et al., 2018](#)). In this context, [Chaboud et al. \(2021\)](#) conduct event studies on EBS and document that the tick size has a significant impact on the importance of limit order activity. The decimalization – i.e. the reduction of tick size by a factor of ten – in the euro-dollar pair in 2011 strongly reduced (increased) the importance of limit order flow at the best quotes (inside the spread). The "half-pip" rule that was implemented in 2012 and partially reversed the decimalization policy lead to an increase in the information content of limit orders submitted at the best quotes. Hence, on Coinbase, price-improving limit order submissions are attractive for informed investors not only due to lower transaction fees but also due to the very low costs of gaining priority over other market participants.

Another factor for the pronounced importance of limit order activity is the structure of the crypto market. With more than 300 spot trading venues and about 40 derivatives exchanges, the crypto market is extremely fragmented.¹⁸ In addition, unlike traditional equity markets, where for example the National Best Bid and Offer (NBBO) regulation of the Securities and Exchange Commission (SEC) ensures that market participants receive the best price when executing trades, the still-unregulated crypto market does not offer a mechanism connecting multiple exchanges. Due to the absence of such a unified framework, there is great potential (and need) for cross-exchange market making and arbitrage activities within the crypto market ([Makarov and Schoar, 2020](#)). Typically, such strategies involve a large number of limit order submissions and cancellations – especially in light of the larger fees for market orders – in order to incorporate information learned from other (faster) exchanges ([Van Kervel, 2015](#); [Shkilko and Sokolov, 2020](#)), leading to the large information content of limit order activity.

A more fundamental possible factor for the importance of limit order activity is related to the works of [Bloomfield et al. \(2005\)](#) and [Rosu \(2020\)](#). They argue that the informed traders' order choice depends on the value of their information. If their privately observed mispricing, defined as the difference between the privately observed fundamental value of the asset and the public expectation of the

¹⁷The exact fee structure can be found on [Coinbase](#).

¹⁸According to [coinmarketcap.com](#).

fundamental value, is large enough, they demand liquidity, otherwise they provide liquidity. However, for cryptocurrencies, there is far no public consensus on their fundamental value. Most recently, [Biais et al. \(2022\)](#) determine the fundamental value of cryptocurrencies as their stream of net transactional benefits, which depend on their future prices. From this, they conclude that cryptocurrencies exhibit multiple equilibria and extrinsic volatility, i.e. prices fluctuate even when fundamentals are constant. On the basis of this, it is far from trivial for informed agents to determine the mispricing and consequently, speculation plays a greater role. However, as shown by [Goettler et al. \(2009\)](#), speculators prefer providing liquidity over demanding it, leading to the high information content of limit order activity.

For limit order submissions, Table 6 shows that price-improving actions are clearly most important. With impacts of ± 0.03 bps, submissions inside the spread contribute about ten times as much as submissions at the best quotes. Behind the best quotes, submissions consistently have absolute price impacts of less than 0.004 bps and in total contribute only about 4% to price discovery. These results are in line with [Brogaard et al. \(2019\)](#) who for Canadian equities, identify orders improving on the NBBO as the most important ones and find limit order activity behind the best quotes contributing very little to price discovery. In comparison to the results by [Chaboud et al. \(2021\)](#) however, the permanent price impact estimates of limit order submissions in Table 6 are of small magnitude. For example, during the last year of their sample, [Chaboud et al. \(2021\)](#) find price-improving limit orders to exhibit price impacts of around 0.075 bps. The reason for this discrepancy is again the tick size. According to [Hautsch and Huang \(2012\)](#), the impact of price-improving limit order submissions increases with relative tick size. Since euro-dollar trading on EBS exhibits a relative tick size of close to one basis point – which is significantly larger than the relative tick size for BTCUSD on Coinbase – an undercutting limit order has a larger price impact for EURUSD on EBS than for BTCUSD on Coinbase.

Interestingly, limit orders submitted more than two dollars away from the best quote have price impacts opposite their order sign. As argued by [Brogaard et al. \(2019\)](#) who also observe this behaviour, this does not imply that an order placed very deep in the book *causes* the mid price to move, but the order rather *predicts* the movement. That is, orders are placed deep in the book (to gain priority over other market participants) when the mid price is about to move in the appropriate direction. For example, when the mid price is about to increase, or already has started to increase, traders might place a sell limit order at a rather high price in order to gain order book priority, which would cause the sell limit order to exhibit a positive price impact. Interestingly, if we partition aggressiveness of limit order submissions more granularly, the results (not presented here) indicate a certain asymmetry in the distance from the best quote at which such ”predicting” orders are placed. On the ask side, we observe a positive price impact for orders submitted two to ten dollars above the best ask quote. On the bid side however, the price impact is negative for all orders submitted more than two dollars below the best bid price. We think that this asymmetry might be related to the leverage effect of volatility that has also been identified for cryptocurrency markets ([Alexander et al., 2021](#)). When the bitcoin price starts to decrease, market participants expect higher volatility, and consequently place their limit orders further away from the best quote, compared to when the price starts to increase.

Finally, we note that apart from very deep in the book, the impact of cancellations behind the best

quotes coincides with their sign, i.e. the cancellation of a buy (sell) limit order is associated with an increase (decrease) in the mid price. When training the RL model containing only cancellations (results not presented here), we do not observe this behaviour. Instead, we find small but significant price impacts opposite to the order sign, that is, cancelling a buy (sell) limit order behind the best quotes leads to an increase (decrease) in the mid price. This indicates that the change in sign of permanent price impacts is caused by the interaction between trades, limit order submissions and cancellations. Due to the recursive architecture of the RL model, the impact of market and limit orders that follow a cancellation are partially attributed to the cancellation, which offsets and overcomes the small impacts of cancellations behind the best quotes. However, similar to submissions, limit order cancellations behind the best quotes exhibit very small absolute price impacts of less than 0.004 bps and despite their abundance, contribute only about 6% to price discovery.

Similar to the previous section, we now investigate how order size affects permanent price impact when all order flow components are considered jointly. To this end, we once more divide all orders into six quantile groups depending on their size.¹⁹ However, in light of the results in Table 6, we only include limit order submissions and cancellations occurring within the spread or at the best quotes. Since activities behind the first level contribute only very little to price discovery, this does not change our results significantly. Nevertheless, we still keep the differentiation of limit order activity from the previous analysis. That is, limit order submissions are stratified based on whether they occur inside the spread (i.e. improve the best quote) or at the best quote (i.e. match the best quote), while cancellations at the best quotes are classified by whether they lead to an immediate change in the mid price (i.e. worsen the best quote) or leave it unchanged (i.e. match the best quote). Consequently, the action space of the RL model includes 60 elements – two order signs and six order size groups for market orders and two order signs, two different immediate mid price reactions, and six order size groups for both limit order submissions and cancellations.

The results of the RL model are reported in Table 7. Overall, they confirm and expand our previous finding on the pronounced importance of limit order activity with regards to information transmission. The permanent price impact of market orders is still significantly larger than that of limit order submissions, regardless of whether the latter improve the best quote. Specifically, depending on the order size, the impact of sell market orders is 2.2 to 3.6 times the impact of a price-improving sell limit order of similar size and even 3.9 to 8.8 times the impact of a price-matching limit order. On the bid side, the discrepancy shows a higher variation, with buy-initiated market orders exhibiting impacts of 1.1 to 3.7 (2.2 to 10.6) times the impact of price-improving (price-matching) limit orders of similar size. However, if we take into account the number of limit order submissions in each order size group, roles are reversed. Across all groups, submissions of price-improving limit orders contribute significantly more to price discovery than market orders. For instance, submissions of very small price-improving buy limit orders (Q1) contribute about 2.9%, whereas buy-initiated market orders of similar size exhibit price discovery contributions of only 1.4%. In some cases, even price-matching submissions at the best quotes contribute more than market orders.

Due to the large price impact of limit order cancellations, that is mostly of mechanical nature,

¹⁹For the sake of consistency, we determine the quantile groups across all components rather than for each one separately.

Table 6 Permanent Price Impact: Market and Limit Orders

The table reports the permanent price impact estimates (in basis points) for market orders (MO), limit order submissions (LO Sub) and limit order cancellations (LO Canc), together with bootstrapped confidence intervals and standard errors, as well as the number of observations for each action. Market orders are differentiated solely on their sign, limit order submissions are distinguished by both their sign and their aggressiveness, and cancellations of limit orders are classified by their sign, their aggressiveness and whether they lead to an immediate change in the mid price (Worsening) or leave it unchanged (Matching). Note that a negative distance to the best same-side quote for limit order submissions indicates that the order improves upon the best same-side quote (Improving). The confidence intervals and standard errors are computed from 100 simulations following the block bootstrapping procedure described in Philip (2020). RC represents the relative contribution of each action to the total amount of price variation (in %), computed as in Equation (3).

Action	DTB	Mid Price	Count	Impact	Lower 2.5%	Upper 2.5%	St. Err.	RC	
Sell	MO	0		792,735	−0.1084	−0.1091	−0.1078	0.0004	8.26
		< 0	Improving	6,414,655	−0.0346	−0.0348	−0.0344	0.0001	21.33
	LO Sub	0	Matching	3,141,497	−0.0034	−0.0034	−0.0034	< 10 ^{−4}	1.03
		0.01		425,090	−0.0030	−0.0031	−0.0030	< 10 ^{−4}	0.12
		0.02−0.05		343,424	−0.0028	−0.0029	−0.0027	< 10 ^{−4}	0.09
		0.05−0.10		319,928	−0.0035	−0.0036	−0.0034	< 10 ^{−4}	0.11
		0.10−0.25		693,542	−0.0035	−0.0035	−0.0034	< 10 ^{−4}	0.23
		0.25−0.50		1,042,044	−0.0029	−0.0029	−0.0028	< 10 ^{−4}	0.29
		0.50−1.00		1,685,528	−0.0020	−0.0021	−0.0020	< 10 ^{−4}	0.33
		1.00−2.00		2,825,264	−0.0010	−0.0010	−0.0010	< 10 ^{−4}	0.28
		≥ 2		21,055,363	0.0002	0.0002	0.0002	< 10 ^{−4}	0.44
		0	Worsening	1,573,868	0.0893	0.0889	0.0897	0.0002	13.51
	LO Canc	0	Matching	1,135,575	< 10 ^{−4}	−0.0001	< 10 ^{−4}	< 10 ^{−4}	0.00
		0.01		1,118,259	−0.0014	−0.0014	−0.0013	< 10 ^{−4}	0.15
		0.02−0.05		1,527,963	−0.0033	−0.0033	−0.0032	< 10 ^{−4}	0.48
		0.05−0.10		755,359	−0.0032	−0.0033	−0.0032	< 10 ^{−4}	0.23
		0.10−0.25		1,129,161	−0.0030	−0.0030	−0.0029	< 10 ^{−4}	0.33
		0.25−0.50		1,447,087	−0.0021	−0.0022	−0.0021	< 10 ^{−4}	0.30
		0.50−1.00		2,259,387	−0.0015	−0.0015	−0.0015	< 10 ^{−4}	0.33
		1.00−2.00		3,659,924	−0.0008	−0.0008	−0.0008	< 10 ^{−4}	0.29
		≥ 2		22,324,451	0.0001	0.0001	0.0001	< 10 ^{−4}	0.19
Buy	MO	0		1,295,979	0.0630	0.0626	0.0635	0.0002	7.86
		< 0	Improving	7,634,526	0.0305	0.0304	0.0307	0.0001	22.40
	LO Sub	0	Matching	4,193,746	0.0032	0.0031	0.0032	< 10 ^{−4}	1.28
		0.01		584,536	0.0028	0.0028	0.0028	< 10 ^{−4}	0.16
		0.02−0.05		469,613	0.0022	0.0021	0.0023	< 10 ^{−4}	0.10
		0.05−0.10		446,759	0.0030	0.0029	0.0031	< 10 ^{−4}	0.13
		0.10−0.25		848,514	0.0030	0.0030	0.0031	< 10 ^{−4}	0.25
		0.25−0.50		1,146,475	0.0024	0.0024	0.0024	< 10 ^{−4}	0.26
		0.50−1.00		1,729,113	0.0015	0.0015	0.0016	< 10 ^{−4}	0.26
		1.00−2.00		2,619,198	0.0007	0.0007	0.0008	< 10 ^{−4}	0.18
		≥ 2		21,798,480	−0.0004	−0.0004	−0.0004	< 10 ^{−4}	0.81
		0	Worsening	1,488,842	−0.0988	−0.0991	−0.0984	0.0002	14.15
	LO Canc	0	Matching	1,316,697	0.0001	0.0001	0.0002	< 10 ^{−4}	0.02
		0.01		1,452,636	0.0019	0.0019	0.0019	< 10 ^{−4}	0.27
		0.02−0.05		2,365,114	0.0038	0.0038	0.0038	< 10 ^{−4}	0.86
		0.05−0.10		1,117,042	0.0035	0.0034	0.0035	< 10 ^{−4}	0.37
		0.10−0.25		1,561,111	0.0033	0.0032	0.0033	< 10 ^{−4}	0.49
		0.25−0.50		1,745,226	0.0022	0.0022	0.0023	< 10 ^{−4}	0.38
		0.50−1.00		2,516,970	0.0015	0.0014	0.0015	< 10 ^{−4}	0.35
		1.00−2.00		3,651,241	0.0007	0.0007	0.0007	< 10 ^{−4}	0.24
		≥ 2		23,303,142	−0.0004	−0.0004	−0.0004	< 10 ^{−4}	0.88

market orders exhibit a smaller impact than price-worsening cancellations across the first three order size groups (Q1–Q3). However, for larger order sizes (Q4–Q6), market orders cause a mid price movement that is 1.2 to 2.5 times larger than that associated with price-worsening cancellations. For example, a very large buy-initiated market order increases the mid price by 0.24 bps, whereas the price-worsening cancellation of a similarly-large sell limit order exhibits a price impact of only 0.1 bps. However, analogous to submissions, price-worsening limit order cancellations are much more frequent, and therefore, apart from the largest order size group (Q6), they contribute significantly more to price discovery than market orders.

Finally, we note that price-matching cancellations of moderate to large size (Q3–Q6) exhibit a permanent price impact whose direction coincides with their order sign, i.e. buy (sell) cancellations are associated with an increase (decrease) in mid price. A close examination of the data shows that these cancellations are remarkably often followed by submissions of price-matching (and to a lesser extent, also price-improving) limit orders on the same order book side and in the same order size group. Due to the recursive architecture of the RL model, this causes the permanent price impact of cancellations to change its sign. Such sequences of cancellations and resubmissions are either driven by queue positioning (traders learn that their order is too far down the queue, decide to cancel the order and submit a new order undercutting the best quote) or order modifications (Coinbase does not allow limit orders to be modified and thus, if traders want to modify the volume of an already submitted limit order, they have to cancel this existing order and submit a new one).

Figure 4 depicts the permanent price impacts of all three order flow components as a function of signed order size and highlights the large differences in price impact between the individual components. The price impact of market orders is still an increasing and concave function of trade size, but compared to Figure 3, the curve appears much smoother now that limit order actions are included in the RL model. For limit order submissions, the functional shape of permanent price impact resembles a sigmoid function, with an apparent limit to which the price impact of limit orders converges. For submissions improving on the best quotes, this limit is around ± 0.07 bps, whereas the impact of submissions at the best quotes seems to converge to roughly ± 0.02 bps. Besides from disproportionately large values for very small order sizes, permanent price impact of price-worsening cancellations, as a function of order size, resembles to some extent a reversed S-shape, similar to a sigmoid function mirrored at the y-axis. Therefore, apart from the reversed direction and the larger magnitude – which is of rather mechanical nature – the price impact of price-worsening limit order cancellations is overall similar to that of limit order submissions undercutting the best quotes.

4.3 Intraday Pattern

In financial markets, information are usually not incorporated into prices at a constant rate and thus, many information-related market variables exhibit a certain intraday variation. Recent studies of equity markets report an S-shaped intraday pattern in bid-ask spreads where the spread is highest after opening and then declines throughout the trading day (Upson and Van Ness, 2017). This intraday shape is usually attributed to index-tracking investors and algorithmic and high-frequency traders who trade extensively at the end of the trading day to rebalance portfolios and to achieve a close-to-zero

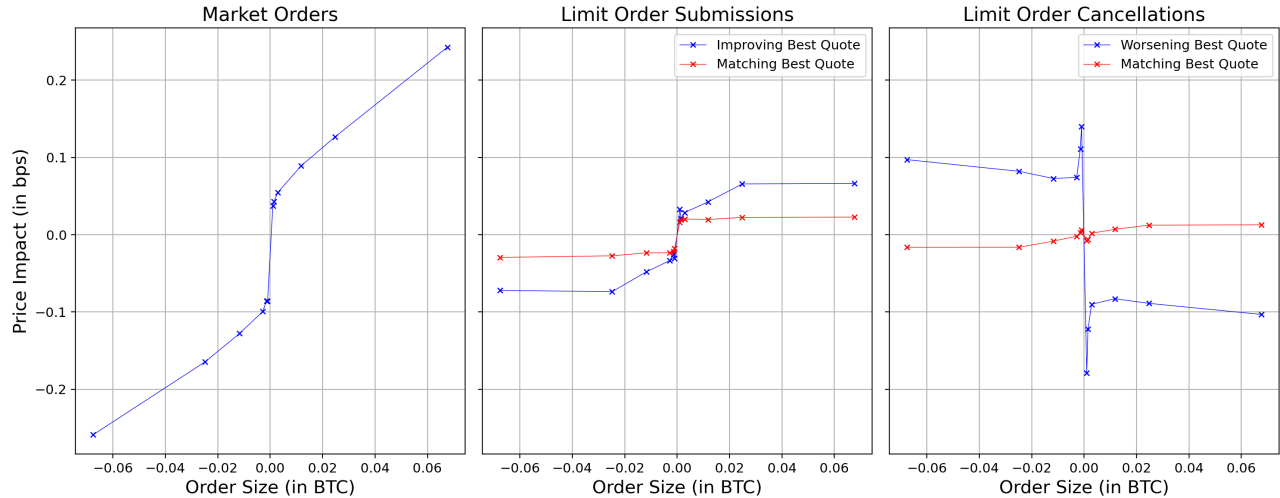
Table 7 Permanent Price Impact: Market and Limit Orders – Order Size

The table reports the permanent price impact estimates (in basis points) for market orders (MO), limit order submissions (LO Sub) and limit order cancellations (LO Canc) at the best quotes, depending on their order size. Market orders are differentiated solely on their sign, limit order submissions and cancellations are distinguished by both their sign and whether they lead to an immediate change in the mid price (Improving, Worsening) or not (Matching). The components are additionally divided into quantile groups (Q1 to Q6) depending on their size. The confidence intervals are computed from 100 simulations following the block bootstrapping procedure described in Philip (2020). RC represents the relative contribution of each action to the total amount of price variation (in %), computed as in Equation (3).

Action	Mid Price	Size	Sell				Buy			
			Impact	Lower 2.5%	Upper 2.5%	RC	Impact	Lower 2.5%	Upper 2.5%	RC
MO		Q1	-0.0858	-0.0867	-0.0848	1.60	0.0377	0.0373	0.0381	1.43
		Q2	-0.0862	-0.0877	-0.0850	0.61	0.0430	0.0424	0.0437	0.57
		Q3	-0.0995	-0.1006	-0.0982	1.03	0.0545	0.0539	0.0550	1.14
		Q4	-0.1281	-0.1296	-0.1267	0.93	0.0890	0.0878	0.0899	0.81
		Q5	-0.1645	-0.1670	-0.1623	0.71	0.1270	0.1250	0.1289	0.69
		Q6	-0.2592	-0.2619	-0.2564	3.33	0.2429	0.2398	0.2457	2.99
LO Sub	Improving	Q1	-0.0305	-0.0308	-0.0303	2.99	0.0331	0.0328	0.0335	2.86
		Q2	-0.0251	-0.0253	-0.0249	1.91	0.0213	0.0212	0.0216	2.84
		Q3	-0.0336	-0.0339	-0.0334	2.36	0.0287	0.0285	0.0290	2.81
		Q4	-0.0481	-0.0484	-0.0477	3.82	0.0425	0.0422	0.0428	3.84
		Q5	-0.0737	-0.0742	-0.0733	6.29	0.0658	0.0653	0.0662	6.45
		Q6	-0.0722	-0.0728	-0.0715	5.81	0.0667	0.0660	0.0672	5.13
	Matching	Q1	-0.0177	-0.0179	-0.0176	0.74	0.0161	0.0159	0.0163	0.67
		Q2	-0.0220	-0.0222	-0.0218	0.81	0.0196	0.0194	0.0198	1.23
		Q3	-0.0234	-0.0237	-0.0232	0.97	0.0205	0.0203	0.0207	1.40
		Q4	-0.0236	-0.0238	-0.0234	0.94	0.0198	0.0196	0.0200	1.01
		Q5	-0.0274	-0.0276	-0.0272	1.01	0.0227	0.0224	0.0229	1.02
		Q6	-0.0295	-0.0298	-0.0293	1.27	0.0230	0.0228	0.0232	1.16
LO Canc	Worsening	Q1	0.1402	0.1394	0.1411	2.17	-0.1790	-0.1800	-0.1781	2.29
		Q2	0.1113	0.1100	0.1124	1.01	-0.1223	-0.1232	-0.1214	1.69
		Q3	0.0743	0.0734	0.0750	1.07	-0.0901	-0.0909	-0.0893	1.23
		Q4	0.0727	0.0721	0.0733	1.75	-0.0828	-0.0833	-0.0822	1.92
		Q5	0.0820	0.0811	0.0829	2.18	-0.0888	-0.0897	-0.0879	2.26
		Q6	0.0972	0.0964	0.0979	2.95	-0.1032	-0.1042	-0.1021	2.54
	Matching	Q1	0.0065	0.0060	0.0068	0.05	-0.0080	-0.0084	-0.0076	0.07
		Q2	0.0030	0.0026	0.0034	0.02	-0.0061	-0.0064	-0.0057	0.06
		Q3	-0.0020	-0.0023	-0.0016	0.03	0.0020	0.0018	0.0022	0.03
		Q4	-0.0083	-0.0086	-0.0081	0.17	0.0071	0.0069	0.0074	0.17
		Q5	-0.0163	-0.0165	-0.0160	0.32	0.0125	0.0123	0.0127	0.28
		Q6	-0.0164	-0.0167	-0.0161	0.31	0.0129	0.0126	0.0132	0.25

Figure 4 Permanent Price Impact of Market and Limit Orders

The figure displays the permanent price impact estimates (in basis points) for market orders (left-hand graph), limit order submissions inside the spread and at the best quotes (middle graph), and limit order cancellations at the best quotes (right-hand graph), as a function of signed order size. Note that sell orders are reported with negative order sizes.



inventory position. In FX markets, which in terms of trading times most closely resemble the crypto market, [Bollerslev and Domowitz \(1993\)](#) document a slightly elevated bid-ask spread during Asian trading hours and a rather low spread in European and US trading. Within the crypto market, [Dimpfl and Maeckle \(2021\)](#) and [Ghysels and Nguyen \(2019\)](#) find a U-shaped intraday pattern in the bid-ask spread for BTCEUR and BTCUSD on Kraken and BTC-e, respectively. More recently, [Jahanshahloo et al. \(2022\)](#) document that trading sessions of the New York Stock Exchange (NYSE) lead bitcoin trading activity, both on the blockchain and centralised exchanges.

To identify a possible intraday pattern in price discovery, we divide the UTC day into 30-minute periods and train the RL model from Table 7 for each intraday period separately. Figure 5 depicts the resulting relative permanent price impacts. That is, for each action, the price impact estimates are first normalised, by dividing by the estimate for the first intraday period, and then averaged across buy and sell. We also include the relative contribution of each 30-minute intraday period to the total price variation. To increase visibility, we only show the price impact estimates for moderately-sized orders of the third quantile size group, but the remaining groups exhibit a very similar pattern. In addition, for limit order activity, only price-changing actions are depicted.²⁰ We see that the intraday pattern is very similar for all three order flow components. During Asian trading in the early UTC morning, price impact is generally rather low. When the London Stock Exchange (LSE) then opens at 07:00 UTC, it jumps up by 20% to 30% and continues to fluctuate around this higher level. At 13:30 UTC, the NYSE opens and we observe a massive jump in price impact during the subsequent 30-minute period. While limit order submissions and cancellations show an increase of 75% and 43%,

²⁰While submissions of price-matching limit orders show a very similar intraday pattern as the actions presented here, price-matching cancellations do not seem to exhibit significant intraday variation.

respectively, the impact of market orders increases by more than 90%. After this peak, price impact shows a few smaller spikes between 16:30 and 18:30 UTC, but overall exhibits a declining trend and reaches its minimum around 22:00, i.e. shortly after closing at the NYSE. The intraday pattern of the relative price discovery contribution coincides with that of permanent price impact. In particular, we see that the NYSE trading periods are clearly most relevant in terms of price discovery. Especially the 30-minute opening period is highly important and contributes by far the most (5%). This intraday pattern confirms the finding of [Jahanshahloo et al. \(2022\)](#) that trading sessions of the NYSE are most relevant in terms of bitcoin trading activity. Moreover, it resembles the M-shaped intraday pattern of volatility that is typically observed for FX pairs ([Andersen and Bollerslev, 1998](#)). This similarity is not entirely surprising since the structures of FX and cryptocurrency markets are similar in that both operate continuously from Monday to Friday and trading is not interrupted overnight.

During the early UTC morning, there is one notable exception to the otherwise quite low price impact. In the 30-minute period from 04:00 to 04:30 UTC, the price impact strongly increases for all three order flow components, with trades showing the largest relative jump of almost 60% compared to the preceding 30-minute interval. This spike is caused by an extremely large trading volume of almost 700 BTC within a ten-minute interval in the early UTC morning of 18 April 2022. During this period, the bitcoin price dropped from almost 40,000 USD to below 38,700 USD (see [Figure 2](#)). Following [Hautsch and Huang \(2012\)](#), this extremely large trading volume, paired with the otherwise comparatively low activity in the market in the early UTC morning, leads to the very high price impact in [Figure 5](#) between 04:00 and 04:30 UTC.

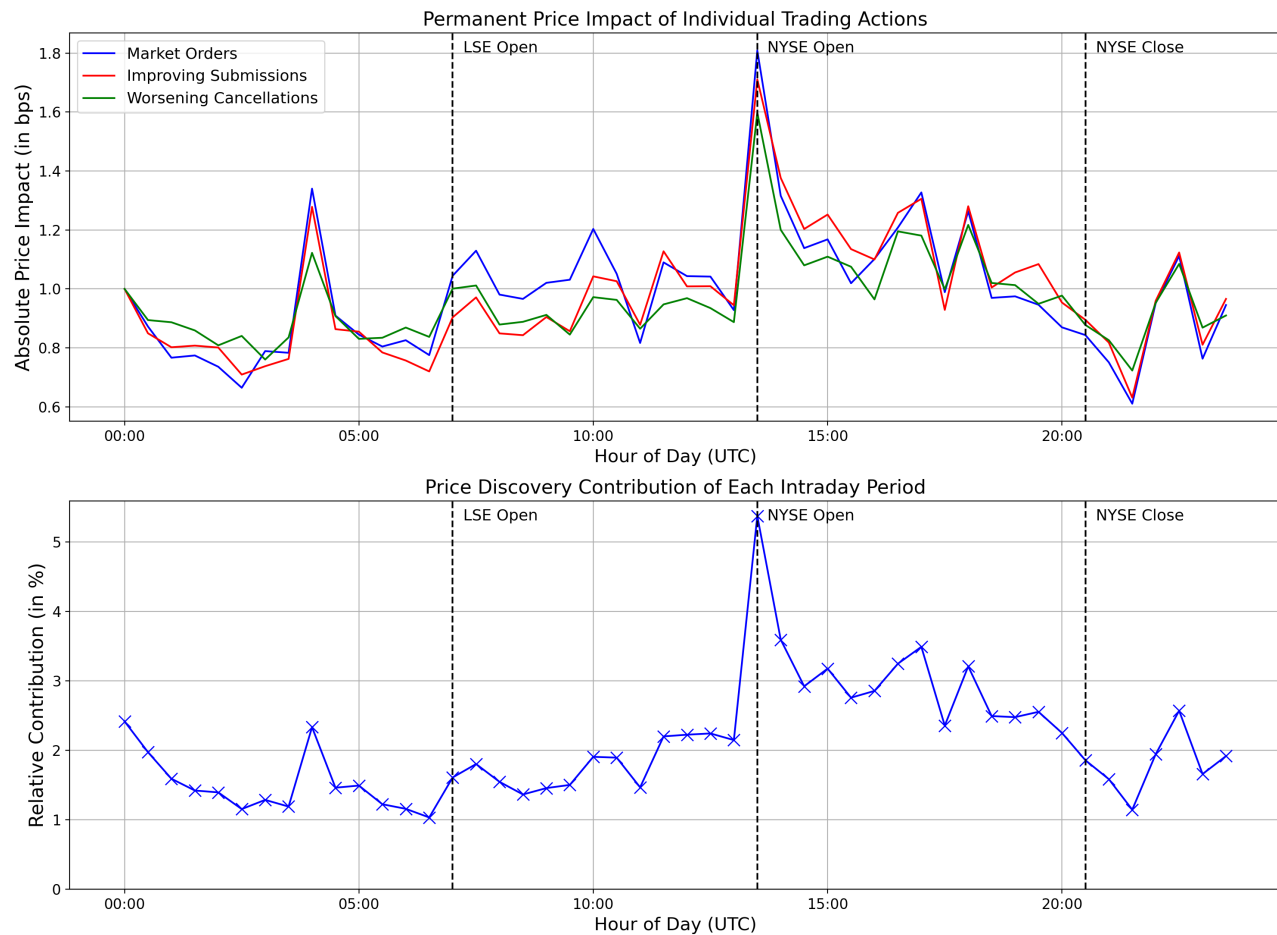
5 The Influence of the Order Book

Throughout the previous section, we have ignored the state of the order book that prevails when the respective action occurs, by assuming that there is only one possible market state (open) in the RL model. This way, we obtained a permanent price impact estimate for each action that is averaged over all possible states of the order book. However, it is well documented that in modern financial markets, the shape of the order book carries a significant amount of information and influences the behaviour of market participants. For example, [Cao et al. \(2009\)](#) document that order imbalances between demand and supply schedules have significant power in predicting short-term returns, even after controlling for autocorrelations in returns and the bid-ask spread. Similarly, [Brogaard et al. \(2014\)](#) find that high-frequency traders use order book information to demand liquidity. Therefore, in this section, we investigate how the permanent price impact of individual actions changes depending on the prevailing order book state.

Formally, we incorporate the state of the order book into the RL model by expanding its state space appropriately. In particular, we no longer assume that there is only one market state (open) but we include different variables into the state space that characterise the shape of the order book. The action space is chosen as in [Table 7](#) and is left unchanged throughout this section. That is, all RL models presented here include all order flow components (market orders, limit order submissions, limit order cancellations) jointly, where for limit order activity, only those orders that affect the best

Figure 5 Intraday Pattern of Price Impact and Price Discovery

The figure displays the intraday pattern of permanent price impact and price discovery during the representative week from 18 to 24 April 2022. The upper graph shows the permanent price impact estimates for moderately-large market orders, price-moving limit order submissions, and price-worsening cancellations. All estimates are normalised, by dividing by the estimate for the first intraday period, and then averaged across buy and sell. The lower graph depicts the relative contribution of each 30-minute intraday period to the total price variation, computed similar to Equation (3).



quotes are included. All order flow components are classified based on their sign and divided into six quantile groups depending on their size.²¹ In addition, limit order submissions are stratified based on whether they occur inside the spread (i.e. improve the best quote) or at the best quote (i.e. match the best quote), while cancellations are classified by whether they lead to an immediate change in the mid price (i.e. worsen the best quote) or leave it unchanged (i.e. match the best quote). Overall, this amounts to 60 different actions in the RL model. Note that all results presented in this section are still based on data for the representative week from 18 to 24 April 2022.

²¹ As before, the quantile groups are computed across market orders, limit order submissions, and limit order cancellations jointly, instead of separately for each order type.

5.1 Order Book Imbalance

Many recent theoretical works such as [Bhattacharya and Saar \(2020\)](#) assume order book imbalance to be informative, with several empirical studies supporting this assumption ([Cao et al., 2009](#); [Gould and Bonart, 2015](#); [Kwan et al., 2021](#)). Within the bitcoin market, [Scaillet et al. \(2020\)](#) document that the aggressive part of the order flow imbalance predicts price jumps. Therefore, it is most likely that the price impact of orders depends significantly on the prevailing order book imbalance. To analyse this, we compute the order book imbalance each time an order is submitted as the normalised difference in cumulative depth between the first five bid and ask price levels, i.e.

$$OBI = \frac{\sum_{i=1}^5 \text{Vol}_i^B - \sum_{i=1}^5 \text{Vol}_i^A}{\sum_{i=1}^5 \text{Vol}_i^B + \sum_{i=1}^5 \text{Vol}_i^A} \quad (4)$$

where Vol_i^B (Vol_i^A) denotes the volume available at the i -th price level on the bid (ask) side of the order book. Due to the normalisation, OBI is between -1 and 1 , where a negative (positive) value implies that the volume on the ask (bid) side outweighs that on the bid (ask) side of the book. Finally, we modify the state space of the RL model by dividing the order book imbalance in five quantile groups ($Q_1^{OBI} - Q_5^{OBI}$) and assign each action to one of the groups depending on the imbalance that prevailed at the time the action occurred. Combined with the action space containing 60 elements, the RL model includes 300 state-action pairs.²²

The resulting permanent price impact estimates and relative price discovery contributions within each state are reported in Table 8.²³ For reasons of clarity, we only include the results on the most negative imbalance group (Q_1^{OBI}) and the most positive one (Q_5^{OBI}) but the results for the three remaining groups are consistent with those presented here. We observe a strong positive relationship between order book imbalance and permanent price impact, with orders submitted or cancelled in the same direction as the order book imbalance leading to much more extreme price impacts than actions in the direction opposite to the imbalance. This relationship is highly robust and holds for all 30 actions. For example, a very large buy market order (Q6) has an impact of only 0.16 bps when the imbalance is most negative (Q_1^{OBI}) compared to 0.34 bps in the case of a high positive book imbalance (Q_5^{OBI}). Similarly, the submission of a rather small price-improving sell limit order (Q2) exhibits an impact of -0.04 bps if the imbalance is most negative, whereas in the case of a highly positive imbalance, it reduces the mid price by only 0.002 bps. For limit order actions that do not induce an immediate change in the mid price, the effect of the imbalance in some cases even offsets and overcomes that of the action, so that the price impact is opposite to the direction of the action. For instance, if the order book is negatively imbalanced, sell limit order cancellations that do not lead to an immediate worsening of the best ask quote exhibit negative price impacts across all order size groups, ranging from -0.04 bps to -0.01 bps.

In general, the relative price discovery contribution of the individual actions within each state is

²²Estimating a VAR of equivalent dimensionality would basically be impossible, since the estimation time of the VAR increases exponentially with the dimensionality of the state space ([Philip, 2020](#)).

²³To prevent clutter, we do not report confidence intervals for the analyses in this section. However, due to the very large sample size, confidence intervals are still quite tight.

also positively related to order book imbalance. Trading actions whose direction coincides with the prevailing imbalance in the order book contribute significantly more to price discovery than orders submitted or cancelled against the order book imbalance. For example, submissions of price-improving buy limit orders overall account for about 33% of price discovery if the order book imbalance is highly positive, whereas they are responsible for only 12% of the total price variation in case of a highly negative imbalance. Within the two most extreme order book imbalance groups (Q_1^{OBI} and Q_5^{OBI}), actions whose direction coincides with the imbalance jointly account for about 70% of price discovery. However, within the less extreme groups (Q_2^{OBI} and Q_4^{OBI}), this dominance is significantly less pronounced and imbalance-conforming actions contribute only slightly more than 50%.

The order book imbalance seems to have no impact on the overall importance of the individual order flow components. The contribution of the components, aggregated over order signs, to the amount of price variation within each state does not change significantly across the different imbalance groups. It is rather the distribution of the price discovery contribution across order signs that changes. For example, the relative contribution of price-improving limit order submissions, aggregated over buy and sell, only varies between 45% and 46% across all five different imbalance groups. If the order book imbalance is highly negative, most of this contribution originates from sell limit orders (34%), while in the case of a high positive imbalance, submissions of buy limit orders are most relevant (33%). Market orders and cancellations of limit orders exhibit a similar pattern. This implies that the order book imbalance does have a significant influence on the permanent price impact of individual actions, however it does not change the order type that informed agents choose to incorporate their information.

Figure 6 depicts the permanent price impact of market orders, limit order submissions and limit order cancellations as a function of their signed order size, for the two most extreme order book imbalance groups (Q_1^{OBI} and Q_5^{OBI}). To increase visibility, we only include price-changing actions for limit order submissions and cancellations. Comparing the graphs to Figure 4, we see that the permanent price impact of market orders is still nonlinear, even after accounting for order book imbalance. However, the imbalance seems to change the functional shape of price impact. The impact of orders that are submitted against the order book imbalance does not increase as strongly in order size as the impact of orders that are submitted with the order book imbalance. That is, the imbalance of the order book appears to dampen the effect of order size for those orders that are submitted against the imbalance. For example, for the most negative order book imbalance group, the impact of buy market orders increases from about 0.01 bps for very small orders to 0.16 bps for the largest orders, which corresponds to an increase of 0.15 bps. However, if the imbalance is most positive, the impact of buy market orders exhibits an increase of 0.28 bps over the same order size range. For limit order submissions and cancellations, including the order book imbalance does not seem to change the functional shape of permanent price impact significantly, but it rather leads to an upward (downward) shift of the curve for positive (negative) imbalances.

5.2 Bid-Ask Spread

Next, we analyse how permanent price impact changes with the bid-ask spread. In an early study of cryptocurrency markets, [Dimpfl \(2017\)](#) finds that the largest part of the spread can be attributed to

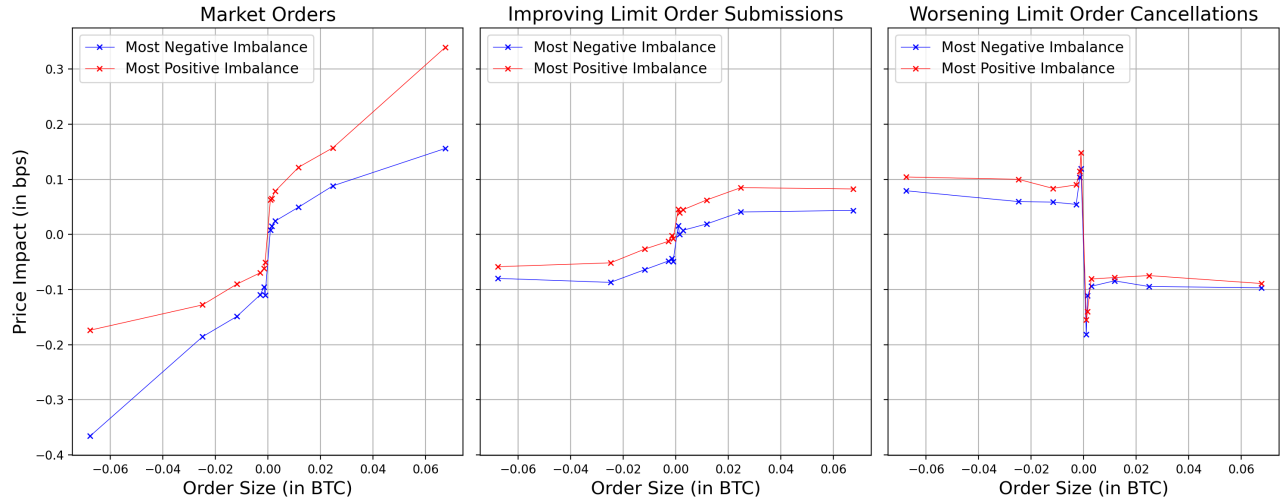
Table 8 Permanent Price Impact: Market and Limit Orders – Order Book Imbalance

The table reports the permanent price impact estimates (in basis points) for market orders (MO), limit order submissions (LO Sub) and limit order cancellations (LO Canc) at the best quotes, depending on the order size and the prevailing order book imbalance. Market orders are differentiated solely on their sign, limit order submissions and cancellations are distinguished by both their sign and whether they lead to an immediate change in the mid price (Improving, Worsening) or leave it unchanged (Matching). The components are additionally divided into quantile groups (Q1 to Q6) depending on their size. The order book imbalance is calculated according to Equation (4) and then divided into five quantile groups ($Q_1^{OBI} - Q_5^{OBI}$). RC_S represents the relative contribution of each action to the amount of price variation within the specific state (in %), computed similar to Equation (3).

Action	Mid Price	Size	Most Negative Imbalance (Q_1^{OBI})				Most Positive Imbalance (Q_5^{OBI})			
			Sell		Buy		Sell		Buy	
			Impact	RC_S	Impact	RC_S	Impact	RC_S	Impact	RC_S
MO		Q1	-0.1110	2.35	0.0079	0.29	-0.0506	0.82	0.0634	2.48
		Q2	-0.0959	0.93	0.0149	0.18	-0.0617	0.31	0.0658	0.96
		Q3	-0.1098	1.47	0.0240	0.48	-0.0691	0.58	0.0789	1.74
		Q4	-0.1493	1.35	0.0490	0.41	-0.0897	0.54	0.1223	1.22
		Q5	-0.1860	0.86	0.0881	0.39	-0.1274	0.45	0.1575	0.90
		Q6	-0.3657	2.85	0.1554	2.14	-0.1733	2.56	0.3398	2.82
LO Sub	Improving	Q1	-0.0491	4.70	0.0157	1.28	-0.0075	0.76	0.0460	4.73
		Q2	-0.0439	2.30	-0.0003	0.05	-0.0020	0.22	0.0396	3.89
		Q3	-0.0486	3.30	0.0073	0.87	-0.0122	1.00	0.0452	3.66
		Q4	-0.0642	5.22	0.0186	2.11	-0.0262	2.24	0.0628	4.67
		Q5	-0.0874	7.63	0.0404	4.62	-0.0513	4.41	0.0853	6.87
		Q6	-0.0801	10.95	0.0433	2.93	-0.0584	3.28	0.0829	9.30
	Matching	Q1	-0.0368	1.25	-0.0007	0.03	0.0013	0.06	0.0338	1.49
		Q2	-0.0403	0.94	0.0009	0.07	-0.0017	0.08	0.0369	1.75
		Q3	-0.0411	1.16	0.0022	0.18	-0.0037	0.20	0.0376	1.84
		Q4	-0.0443	1.61	0.0012	0.06	-0.0023	0.09	0.0394	1.79
		Q5	-0.0478	1.56	0.0047	0.22	-0.0064	0.23	0.0422	1.66
		Q6	-0.0485	2.46	0.0096	0.49	-0.0126	0.46	0.0427	2.24
LO Canc	Worsening	Q1	0.1190	1.56	-0.1820	3.09	0.1488	2.84	-0.1542	1.79
		Q2	0.1031	0.60	-0.1116	2.87	0.1149	1.83	-0.1396	1.10
		Q3	0.0543	0.70	-0.0945	2.17	0.0904	1.73	-0.0804	0.81
		Q4	0.0582	1.30	-0.0845	3.09	0.0840	2.44	-0.0780	1.33
		Q5	0.0592	1.48	-0.0948	3.40	0.1004	3.06	-0.0741	1.43
		Q6	0.0788	4.53	-0.0970	1.39	0.1046	1.42	-0.0887	3.64
	Matching	Q1	-0.0146	0.09	-0.0319	0.22	0.0293	0.26	0.0145	0.13
		Q2	-0.0162	0.06	-0.0296	0.37	0.0272	0.26	0.0158	0.12
		Q3	-0.0193	0.19	-0.0220	0.40	0.0205	0.34	0.0233	0.33
		Q4	-0.0295	0.49	-0.0181	0.39	0.0174	0.37	0.0289	0.58
		Q5	-0.0397	0.70	-0.0149	0.26	0.0141	0.24	0.0359	0.74
		Q6	-0.0321	0.83	-0.0123	0.11	0.0108	0.10	0.0297	0.80

Figure 6 Permanent Price Impact Depending on Order Book Imbalance

The figure displays the permanent price impact estimates (in basis points) of market orders (left-hand graph), price-improving limit order submissions (middle graph) and price-worsening limit order cancellations (left-hand graph) for the two most extreme order book imbalance groups (Q_1^{OBI} and Q_5^{OBI}), as a function of signed order size.



adverse selection. Similarly, [Scaillet et al. \(2020\)](#) report that a widening of the bid-ask spread predicts jumps in the bitcoin price. Therefore, it seems likely that the spread on Coinbase contains a significant amount of information and has a non-trivial influence on permanent price impact.

Similar to order book imbalance, we compute the bid-ask spread each time an action occurs and modify the state space of the RL model by partitioning the spread. However, according to Table 2, the spread is at its minimum value for 37% of the time and its distribution is quite heavily right-skewed. Therefore, we refrain from using quantile groups for partitioning and instead, following [Kwan et al. \(2021\)](#), simply divide the spread into two states: binding (0.01 USD) and wide (>0.01 USD). Each action is then assigned to one of the two states depending on the bid-ask spread that prevailed at the time the action occurred. Since price-improving limit order submissions cannot occur if the spread is binding, this partitioning leads to only 168 state-action pairs in the RL model.

Table 9 reports the resulting permanent price impact estimates and relative price discovery contributions within each state. We see that for both buy and sell, market orders generally exhibit a significantly higher price impact, and thus carry more information, if the bid-ask spread is binding. For instance, a very small buy market order increases the mid price by 0.06 bps in case of a binding spread, compared to only 0.03 bps if the spread is wide. Moreover, the overall contribution of market orders to price discovery is significantly higher in the case of a binding spread (38%) than if the spread is wide (11%). This finding is in line with the prediction of [Bhattacharya and Saar \(2020\)](#) that informed agents submit market orders when the spread is narrow.

The only exception to this are very large market orders in the top quantile size group, which contain more information in case of a wide spread. Since a widening of the spread is typically an indicator for an imminent increase in volatility ([Scaillet et al., 2020](#)), we suppose that these large orders are submitted by agents that anticipate such an increase in volatility and consume the liquidity

available on the opposite side of the book before it vanishes. To confirm this hypothesis, we re-train the RL model with an additional state in which the spread is larger than two dollars, corresponding approximately to its 60%-quantile. If the high price impact of very large orders was due to agents taking liquidity in the anticipation of elevated volatility, the impact should even be more extreme in the newly-introduced widest-spread state. Indeed, the results confirm this hypothesis. While the impact of small to moderately-large market orders (Q1 to Q5) reduces as the spread increases, very large orders in the top quantile group (Q6) exhibit price impacts that consistently increase across all three spread states. This finding is in contrast to [Kwan et al. \(2021\)](#) who document that market orders have a larger price impact across all order size groups in case of a binding spread. The discrepancy is probably related to the high volatility of cryptocurrencies compared to equities. Since large price swings occur much more frequently for bitcoin than for equity, we observe such large volatility-anticipating trades significantly more often, which leads to the larger price impacts in case of a wide spread. However, these large trades at a wide bid-ask spread still tend to occur comparatively rarely and therefore, despite the larger individual price impact in case of a wide spread, their relative contribution to price discovery is still higher when the spread is binding (6.68% for sell trades and 6.24% for buy transactions) than when it is wide (2.54% and 2.25%).

For limit order submissions, results are more nuanced. In case of a wide spread, price-improving submissions are highly important and in total account for more than 57% of price discovery, which confirms the prediction of [Bhattacharya and Saar \(2020\)](#) that informed traders rely on limit orders when the bid-ask spread is wide. If the spread is binding, price-improving limit orders cannot be submitted and consequently, submissions at the best quotes become more relevant. However, the large information content of price-improving orders in the case of a binding spread is not entirely transferred to submissions at the best quotes, but it is split and transferred to both market orders and submissions at the best quotes, with the contribution of the latter increasing from below 10% in case of a wide spread to more than 28% for a binding spread.

These results imply that if the bid-ask spread becomes binding, informed agents partially change the type of order they use to incorporate information. While some agents continue to provide liquidity (at the best quotes), others switch to taking it which makes market orders the main driver of price discovery if the bid-ask spread is binding. An obvious explanation for this behaviour are execution costs. In case of a binding spread which in relative terms corresponds to only about 0.0025 bps, market orders (and the associated immediacy) are relatively inexpensive and the value of the agents' information does not need to be as high in order for the profit to exceed execution costs. Another possible explanation is competition among informed market participants in the spirit of [Baruch et al. \(2017\)](#). Informed agents might interpret a binding bid-ask spread as a signal for the presence of other informed traders and therefore, prefer to act quickly on their information by submitting market orders (and paying higher execution costs) rather than facing execution risk (and reducing transaction costs) through limit orders.

Finally, as in previous analyses, price-worsening limit order cancellations have a very high permanent price impact and contribute a significant amount to price discovery. However, a close examination of the data shows that in both cases – binding spread and wide spread – most of this high impact is

of mechanical nature due to large price gaps behind the best quotes that are caused by the extremely small relative tick size. For example, the price-worsening cancellation of a very small sell limit order (Q1) causes an immediate increase of 0.2 bps in the mid price, which is already more than 90% of the entire permanent price impact of 0.22 bps documented in Table 9.

Table 9 Permanent Price Impact: Market and Limit Orders – Bid-Ask Spread

The table reports the permanent price impact estimates (in basis points) for market orders (MO), limit order submissions (LO Sub) and limit order cancellations (LO Canc) at the best quotes, depending on the order size and the prevailing bid-ask spread. Market orders are differentiated solely on their sign, limit order submissions and cancellations are distinguished by both their sign and whether they lead to an immediate change in the mid price (Improving, Worsening) or leave it unchanged (Matching). The components are additionally divided into quantile groups (Q1 to Q6) depending on their size. The bid-ask spread is divided in two states: binding (0.01 USD) and wide (>0.01 USD). RC_S represents the relative contribution of each action to the amount of price variation within the specific state (in %), computed similar to Equation (3).

Action	Mid Price	Size	Binding Spread				Wide Spread			
			Sell		Buy		Sell		Buy	
			Impact	RC_S	Impact	RC_S	Impact	RC_S	Impact	RC_S
MO		Q1	-0.1225	4.94	0.0552	4.31	-0.0614	0.84	0.0271	0.78
		Q2	-0.1080	1.55	0.0547	1.44	-0.0741	0.40	0.0362	0.37
		Q3	-0.1200	2.57	0.0652	2.69	-0.0869	0.68	0.0482	0.78
		Q4	-0.1427	2.21	0.1021	1.93	-0.1184	0.63	0.0811	0.56
		Q5	-0.1701	1.52	0.1310	1.61	-0.1606	0.53	0.1249	0.48
		Q6	-0.2302	6.68	0.2212	6.24	-0.2791	2.54	0.2596	2.25
LO Sub	Improving	Q1					-0.0300	3.56	0.0336	3.52
		Q2					-0.0241	2.22	0.0214	3.46
		Q3					-0.0329	2.80	0.0290	3.45
		Q4					-0.0475	4.59	0.0427	4.69
		Q5					-0.0739	7.65	0.0663	7.89
		Q6					-0.0721	7.05	0.0674	6.29
	Matching	Q1	-0.0233	1.47	0.0124	0.73	-0.0176	0.64	0.0180	0.68
		Q2	-0.0299	1.72	0.0185	1.50	-0.0212	0.68	0.0207	1.21
		Q3	-0.0318	2.54	0.0210	2.11	-0.0225	0.73	0.0213	1.30
		Q4	-0.0326	3.35	0.0208	2.38	-0.0222	0.57	0.0209	0.77
		Q5	-0.0361	3.31	0.0245	2.47	-0.0265	0.65	0.0234	0.75
		Q6	-0.0404	3.93	0.0260	2.67	-0.0274	0.83	0.0230	0.89
LO Canc	Worsening	Q1	0.2236	5.02	-0.2587	5.54	0.1073	1.47	-0.1388	1.49
		Q2	0.1916	1.67	-0.2087	2.66	0.0928	0.85	-0.1021	1.42
		Q3	0.1509	1.16	-0.1805	1.45	0.0650	1.02	-0.0769	1.14
		Q4	0.1423	1.58	-0.1563	1.66	0.0657	1.76	-0.0741	1.91
		Q5	0.2089	2.05	-0.2065	1.78	0.0720	2.17	-0.0790	2.29
		Q6	0.2074	2.58	-0.2280	2.40	0.0880	3.01	-0.0910	2.51
	Matching	Q1	0.0057	0.14	-0.0126	0.33	0.0079	0.03	-0.0080	0.03
		Q2	0.0000	0.00	-0.0099	0.27	0.0061	0.02	-0.0064	0.04
		Q3	-0.0081	0.33	0.0018	0.09	0.0032	0.02	0.0003	0.00
		Q4	-0.0106	0.63	0.0028	0.18	-0.0080	0.09	0.0095	0.13
		Q5	-0.0150	0.79	0.0054	0.31	-0.0204	0.24	0.0178	0.25
		Q6	-0.0195	1.07	0.0085	0.44	-0.0178	0.19	0.0169	0.20

6 Inter-Exchange Information Flows

Due to the high fragmentation of cryptocurrency markets, it is likely that trading on Coinbase is influenced by information from other exchanges. In fact, empirical studies such as Alexander and

Heck (2020) report that spot exchanges, including Coinbase, do not contribute much to price discovery and it is the highly-leveraged derivatives products offered by many unregulated crypto exchanges that dominate the price discovery process. Therefore, in this section, we study how exactly information from other exchanges influences the price discovery process on Coinbase. As before, we consider only actions affecting the best quotes, which according to Table 6, should not change results significantly. Therefore, in the following, the term *event* refers to all messages on Coinbase affecting the best quotes.

To analyse how the permanent price impact on Coinbase is influenced by other exchanges, we obtain tick data on major derivatives instruments and spot pairs from Tardis.dev. More specifically, we retrieve tick data on the top-of-book quotes of the three most actively traded bitcoin derivatives (USDT-perpetual contracts traded on Binance and Bybit and the USD-perpetual offered by FTX) and the spot pair with the highest trading volume (BTCUSDT on Binance). As a sort of control variable, we also obtain quotes data for bitcoin-dollar trading on Bitstamp, a well-established but rather small spot exchange that is likely to have little to no influence on Coinbase. It is important to note that Tardis operates a synchronized clock to timestamp messages upon arrival on their servers. This mitigates the issue of small time differences in the timestamps provided by the exchanges and allows us to replay the market from the perspective of a trader.

For each of these five products (called transmitting products in the following), we now train an RL model where the state space captures the past return on both Coinbase and the product. More specifically, we apply the following procedure for each product. First, for each event t on Coinbase, we determine the as-of mid price of the transmitting product, i.e. the most recent mid price at the time of the Coinbase event. Then, we use these as-of mid prices to calculate the log return of the product over the previous 50 Coinbase events (i.e. $t - 50, \dots, t - 1$), which on average corresponds to a time period of about one second.²⁴ Next, we quantize these mid returns and divide them into five quantile groups ($Q_1^P - Q_5^P$). In the same way, we calculate the 50-message log returns of the mid price on Coinbase and quantize them, yielding the groups $Q_1^{CB} - Q_5^{CB}$. This way, we obtain for each event t on Coinbase, the quantile group of the preceding return on the transmitting product as well as the quantile group of the preceding return on Coinbase. The state space of the RL model is now defined as the set of all 25 possible combinations of the returns on Coinbase and the transmitting product, i.e. $S = \{(Q_i^{CB}, Q_j^P) \mid i, j = 1, \dots, 5\}$. The action space is chosen similar to Table 7 but to not overload the model, we do not differentiate orders based on their size. Therefore, the action space includes only ten actions – buy and sell market order, price-improving and price-matching submissions of buy and sell limit orders, price-worsening and price-matching cancellations of buy and sell limit orders. Consequently, this specification of the RL model includes 250 state-action pairs.

Table 10 provides the permanent price impact estimates from these RL models.²⁵ For reasons of brevity, we only report the results on sell actions for two of the five transmitting products (Binance

²⁴Due to their fully electronic nature, we expect information to be transmitted rather quickly between cryptocurrency exchanges. Even retail traders can easily apply optimised out-of-the-box trading bots that allow them to monitor several exchanges and execute trading strategies based on inter-exchange price differences (see for example the open-source Hummingbot). As a robustness check, we have repeated the procedure with returns calculated over both shorter and longer intervals. However, the results are qualitatively similar to those for 50-event returns.

²⁵As before, we do not report confidence intervals to prevent clutter. However, due to the large sample size, the intervals are still quite tight.

USDT-perpetual and BTCUSD on Bitstamp). However, the results for buy actions and the remaining three products are very similar to those presented here. Moreover, the relative price discovery contributions, calculated using Equation (3), do not yield any further insights and are consistent with the permanent price impacts (i.e. the higher the absolute price impact of an action, the higher its price discovery contribution). Therefore, we refrain from reporting the contributions here.

We see that the past return on the Binance perpetual significantly influences permanent price impacts on Coinbase. If the price on the Binance perpetual has declined over the last 50 events, Coinbase follows and the permanent price impacts of all actions are shifted downwards. On the other hand, if the Binance perpetual has shown a price increase, the impact of all actions on Coinbase increases as well. For instance, a sell market order that is submitted on Coinbase after a negative return on the Binance perpetual (Q_2^P) reduces the mid price by 0.25 bps to 0.37 bps. If the same order is submitted after a moderate price increase on Binance (Q_4^P), the mid price on Coinbase is only reduced at most 0.05 bps. Interestingly, in the case of a highly positive price movement on the Binance perpetual (Q_5^P) the permanent price impact of all actions on Coinbase is positive. For example, the submission of a sell market order on Coinbase after a highly positive return on Binance leads to an increase in the Coinbase mid price of 0.34 bps to 0.75 bps. This implies that the influence of the Binance USDT-perpetual is strong enough to offset and overcome the negative immediate impact of sell actions on Coinbase.

The influence of the Binance perpetual contract on Coinbase is so strong that its returns even affect the permanent price impact of all actions more than the returns on Coinbase itself. If we hold the quantile return group of Coinbase constant, the influence of past Binance returns, measured as the difference in permanent price impact between the most negative and the most positive return group (Q_1^P and Q_5^P), is between 1.38 bps and 1.66 bps. On the other hand, if we hold the quantile return group of Binance constant, the influence of past Coinbase returns ranges only from 0.12 bps to 0.41 bps. This massive difference highlights the dominance of the Binance USDT-perpetual over Coinbase in terms of price discovery and implies that even at such short time frames, the price discovery process on Coinbase is largely dictated by the Binance perpetual.

For Bitstamp, Table 10 shows a different picture. The permanent price impacts on Coinbase are affected very little by the preceding return on Bitstamp. In fact, they are even slightly reduced if there has been a positive return on Bitstamp.²⁶ For instance, a sell market order on Coinbase reduces the mid price by at most 0.33 bps if the preceding return on Bitstamp has been most negative (Q_1^P) while the same order causes the mid price to decrease by at most 0.42 bps in case of a positive price movement on Bitstamp. In contrast to Binance, the influence of Bitstamp on Coinbase is not strong enough to offset and overcome the negative immediate impact of sell actions, so that in case of a highly positive return on Bitstamp (Q_5^P), all sell actions generally still have a negative permanent price impact.²⁷

²⁶Such a reduction could for example arise from a counter-cyclical behaviour of Bitstamp and Coinbase. That is, if Bitstamp follows Coinbase with a time lag, it might be that just after a positive return has been transmitted from Coinbase to Bitstamp, the price on the latter starts to increase, while the price on Coinbase has already lost its momentum and starts falling again.

²⁷In some cases, submissions of sell orders exhibit a positive permanent impact for a highly positive Bitstamp return. However, this behaviour occurs only after a positive return on Coinbase (Q_4^{CB} and Q_4^{CB}) and should therefore not be

Table 10 Permanent Price Impact: Market and Limit Orders – Inter-Exchange Information Flows

The table reports the permanent price impact estimates (in basis points) for market orders (MO), limit order submissions (LO Sub) and limit order cancellations (LO Canc) at the best quotes, depending on the preceding return on both Coinbase and the Binance USDT-perpetual (Bitstamp BTCUSD). Market orders are differentiated solely on their sign, limit order submissions and cancellations are distinguished by both their sign and whether they lead to an immediate change in the mid price (Improving, Worsening) or leave it unchanged (Matching). The preceding returns are calculated as the log change in the mid price over the last 50 events on Coinbase. Then, they are divided into five quantile groups, giving the Coinbaser return groups (Q_1^{CB} to Q_5^{CB}) and the return groups (Q_1^P – Q_5^P) of the Binance USDT-perpetual (Bitstamp BTCUSD). For reasons of brevity, we only report results for sell actions. $Q_5^P - Q_1^P$ reports the difference in permanent price impact between the most negative and the most positive return group of the Binance USDT-perpetual (Bitstamp BTCUSD). $Q_5^{CB} - Q_1^{CB}$ reports the difference in permanent price impact between the most negative and the most positive return group of Coinbase.

Action	Mid Price	Coinbase	Binance USDT-Perpetual						Bitstamp BTCUSD					
			Q_1^P	Q_2^P	Q_3^P	Q_4^P	Q_5^P	$Q_5^P - Q_1^P$	Q_1^P	Q_2^P	Q_3^P	Q_4^P	Q_5^P	$Q_5^P - Q_1^P$
MO		Q_1^{CB}	-1.1150	-0.3718	-0.1652	-0.0459	0.3408	1.4558	-0.3280	-0.3372	-0.3573	-0.4004	-0.4190	-0.0909
		Q_2^{CB}	-0.9996	-0.3143	-0.1179	0.0331	0.4887	1.4883	-0.1856	-0.1833	-0.2036	-0.2061	-0.2809	-0.0953
		Q_3^{CB}	-0.8914	-0.2634	-0.0567	0.0981	0.5924	1.4838	-0.0839	-0.0951	-0.0861	-0.0894	-0.1698	-0.0859
		Q_4^{CB}	-0.8755	-0.2382	-0.0297	0.1267	0.6578	1.5333	-0.0250	-0.0485	-0.0298	-0.0387	-0.1160	-0.0910
		Q_5^{CB}	-0.9168	-0.2506	-0.0155	0.1481	0.7471	1.6638	0.0205	0.0070	0.0613	0.0477	-0.0259	-0.0464
		$Q_5^{CB} - Q_1^{CB}$	0.1982	0.1212	0.1497	0.1940	0.4063		0.3485	0.3442	0.4185	0.4481	0.3931	
LO Sub	Improving	Q_1^{CB}	-0.9697	-0.2948	-0.1064	0.0195	0.4876	1.4573	-0.2018	-0.1961	-0.2410	-0.2701	-0.2787	-0.0769
		Q_2^{CB}	-0.8488	-0.2331	-0.0545	0.0929	0.5686	1.4174	-0.0681	-0.0693	-0.1044	-0.1123	-0.1658	-0.0978
		Q_3^{CB}	-0.7676	-0.1908	-0.0110	0.1412	0.6377	1.4053	0.0068	-0.0095	-0.0236	-0.0315	-0.0937	-0.1005
		Q_4^{CB}	-0.7296	-0.1693	0.0118	0.1701	0.7072	1.4369	0.0559	0.0273	0.0287	0.0164	-0.0388	-0.0947
		Q_5^{CB}	-0.7437	-0.1522	0.0396	0.2127	0.8061	1.5498	0.1134	0.0940	0.1190	0.1134	0.0565	-0.0569
		$Q_5^{CB} - Q_1^{CB}$	0.2260	0.1425	0.1460	0.1932	0.3186		0.3152	0.2901	0.3600	0.3835	0.3352	
	Matching	Q_1^{CB}	-0.9232	-0.2784	-0.0909	0.0329	0.5192	1.4424	-0.1643	-0.1629	-0.2065	-0.2404	-0.2428	-0.0784
		Q_2^{CB}	-0.8190	-0.2235	-0.0448	0.1011	0.5786	1.3976	-0.0478	-0.0542	-0.0881	-0.0974	-0.1456	-0.0978
		Q_3^{CB}	-0.7343	-0.1817	-0.0028	0.1475	0.6484	1.3826	0.0240	0.0024	-0.0108	-0.0218	-0.0738	-0.0978
		Q_4^{CB}	-0.6889	-0.1504	0.0316	0.1869	0.7303	1.4192	0.0827	0.0495	0.0526	0.0357	-0.0123	-0.0950
		Q_5^{CB}	-0.6703	-0.1174	0.0738	0.2480	0.8558	1.5261	0.1668	0.1424	0.1654	0.1596	0.1042	-0.0626
		$Q_5^{CB} - Q_1^{CB}$	0.2529	0.1609	0.1646	0.2151	0.3366		0.3311	0.3052	0.3719	0.4000	0.3470	
LO Canc	Worsening	Q_1^{CB}	-0.7559	-0.1485	0.0341	0.1719	0.7567	1.5125	-0.0048	-0.0149	-0.0594	-0.0848	-0.0760	-0.0712
		Q_2^{CB}	-0.6807	-0.1270	0.0551	0.1989	0.7638	1.4445	0.0811	0.0457	0.0279	0.0070	-0.0199	-0.1011
		Q_3^{CB}	-0.6256	-0.1054	0.0713	0.2233	0.8240	1.4495	0.1403	0.0834	0.0757	0.0573	0.0387	-0.1016
		Q_4^{CB}	-0.5689	-0.0591	0.1224	0.2801	0.9192	1.4881	0.2113	0.1587	0.1711	0.1435	0.1210	-0.0903
		Q_5^{CB}	-0.4916	-0.0106	0.1753	0.3580	1.0384	1.5300	0.3252	0.2890	0.3133	0.3024	0.2609	-0.0643
		$Q_5^{CB} - Q_1^{CB}$	0.2642	0.1379	0.1412	0.1861	0.2817		0.3300	0.3039	0.3727	0.3872	0.3369	
	Matching	Q_1^{CB}	-0.8986	-0.2607	-0.0776	0.0490	0.5660	1.4646	-0.1439	-0.1449	-0.1886	-0.2199	-0.2211	-0.0772
		Q_2^{CB}	-0.7941	-0.2114	-0.0330	0.1151	0.6105	1.4046	-0.0308	-0.0413	-0.0739	-0.0812	-0.1286	-0.0978
		Q_3^{CB}	-0.7140	-0.1731	0.0059	0.1588	0.6694	1.3834	0.0361	0.0114	-0.0010	-0.0100	-0.0594	-0.0955
		Q_4^{CB}	-0.6656	-0.1351	0.0455	0.2052	0.7601	1.4257	0.1021	0.0652	0.0688	0.0547	0.0070	-0.0951
		Q_5^{CB}	-0.6380	-0.0958	0.0932	0.2724	0.8923	1.5303	0.1938	0.1673	0.1902	0.1878	0.1308	-0.0630
		$Q_5^{CB} - Q_1^{CB}$	0.2606	0.1648	0.1708	0.2235	0.3263		0.3377	0.3122	0.3788	0.4077	0.3519	

Overall, the influence of Bitstamp on Coinbase is at best rather weak. If we hold the quantile return group of Coinbase fixed, the influence of past Bitstamp returns, again measured as the difference in permanent price impact between the most negative and the most positive return group, is very small and even negative, ranging from -0.1 bps to -0.05 bps. In contrast, the influence of past Coinbase returns is between 0.31 bps and 0.41 bps. This discrepancy implies that Bitstamp is rather irrelevant with regards to price discovery on Coinbase.

Figure 7 depicts the permanent price impacts of market orders on Coinbase, depending on the preceding return on both Coinbase and the five transmitting products. Note that the respective figures look very similar for submissions and cancellations of limit orders and they are therefore not shown here. The price impact estimates are depicted as heatmaps, i.e. the darker the colour of a cell, the lower the permanent price impact for the respective combination of quantile return groups. For the three perpetual contracts and the BTCUSDT spot pair on Binance, cells in the same column are of very similar colour and most variation is between, rather than within, columns. This implies that the estimates of permanent price impact do not vary much across the different return groups of Coinbase ($Q_1^{CB} - Q_5^{CB}$), but they change significantly depending on the quantile return group of the three perpetuals and the Binance spot pair ($Q_1^P - Q_5^P$). Therefore, we conclude that even at short time frames of about one second, permanent price impacts, and thus the price discovery process, on Coinbase strongly depend on the preceding price movements on other, leading products and exchanges. In contrast, the heatmap for Bitstamp shows a large degree of variation within, rather than across, columns, which indicates that past returns on Coinbase itself are more relevant for the price discovery process on Coinbase than returns on Bitstamp. That is, Coinbase does not follow the price movements on Bitstamp, but it is rather the other way around.

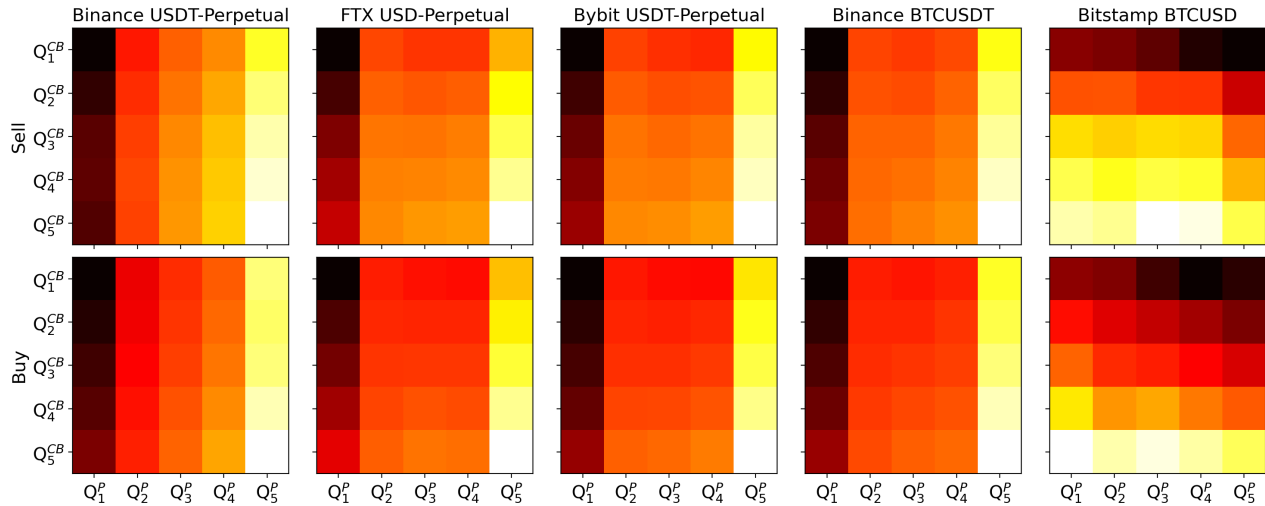
7 Conclusion

Similar to traditional stock exchanges, cryptocurrency exchanges operate a central limit order book and thus, the traditional assumption that only trades carry information is not reasonable any more. Therefore, using an extensive four-month tick-by-tick data set on transactions and order book messages on Coinbase, the most established cryptocurrency spot exchange, we are the first to study the price discovery process of bitcoin-dollar trading at its most granular level. To do so, we apply a recently proposed modified reinforcement learning framework that is able to capture nonlinear relationships and can be trained within a reasonable amount of time, even if tens of millions of observations are included and the variables space is high-dimensional. We estimate the permanent price impacts and the relative price discovery contributions of different order flow components (market orders, limit order submissions, limit order cancellations) and analyse how the state of the order book affects the contribution of individual components. Finally, we examine how other cryptocurrency (derivatives) exchanges influence the price discovery process on Coinbase.

We show that market orders individually exhibit a much larger permanent price impact than limit order submissions, even in case the latter undercut the best quote. However, if the number of attributed to the positive Bitstamp return.

Figure 7 Permanent Price Impact Depending on Inter-Exchange Flows

The figure displays the permanent price impact estimates for market orders on Coinbase, depending on the preceding returns on both Coinbase and one of the five transmitting products (Binance USDT-perpetual, FTX USD-perpetual, Bybit USD-perpetual, BTCUSDT on Binance, BTCUSD on Bitstamp). The upper row of graphs represents sell-initiated market orders, the lower row is for buy-initiated market orders.



occurrences is taken into account, limit order submissions are significantly more important than market orders and contribute about 50% to the price discovery process. The bulk of this large contribution originates from limit order submissions that undercut the best quotes. These findings are consistent with recent studies on FX and equity markets (Brogaard et al., 2019; Chaboud et al., 2021). In contrast to traditional asset classes, we observe a certain asymmetry in the permanent price impact of market orders, with sell-initiated market orders carrying significantly more information than buy orders. We suppose that this discrepancy is caused by unidirectional retail order flows on Coinbase, where retail investors mainly position themselves on the buy side of transactions.

Moreover, we show that the order book has a significant impact on the price discovery process. Orders that are submitted with the order book imbalance contribute considerably more to price discovery than orders whose direction is opposite to the prevailing imbalance. The bid-ask spread has a non-trivial effect on the order choice of informed agents. In case of a non-binding spread, price discovery mainly occurs through undercutting limit orders, whereas if the spread is binding, market orders are most important. Finally, we show that the three most actively traded bitcoin derivatives products and the largest spot pair heavily influence the price discovery process on Coinbase. Trading actions whose direction coincides with the preceding price evolution on the four leading products are the main driver of price discovery. Compared to the returns on the three derivatives products and the largest spot pair, the preceding price evolution on Coinbase itself is rather irrelevant.

With several hundred spot exchanges and a few dozen derivatives trading venues, the cryptocurrency is highly fragmented. However, fragmented markets are still interconnected. As shown by Van Kervel (2015), transactions on one exchange trigger reactive submissions and cancellations of limit orders on other exchanges. Such reactions are efficient and in a price impact analysis, they

should be attributed to the initial transaction. Therefore, in terms of futures research, it would be of great interest (i) to analyse the price discovery process on exchanges other than Coinbase – most importantly the derivatives venues – and (ii) to expand the RL framework used in this study to include several exchanges so that it yields a market-wide permanent price impact estimate, where, for example, cancellations on one exchange that are caused by a transaction on another exchange are correctly assigned to the transaction.

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